



Siemens
Industry
Online
Support

MANUAL

IEC 61850 Client Library TIA Portal

User Manual

SIEMENS

Legal information

Warning notice system

This manual contains notices you have to observe in order to ensure your personal safety, as well as to prevent damage to property. The notices referring to your personal safety are highlighted in the manual by a safety alert symbol, notices referring only to property damage have no safety alert symbol. These notices shown below are graded according to the degree of danger.



DANGER

indicates that death or severe personal injury will result if proper precautions are not taken.



WARNING

indicates that death or severe personal injury may result if proper precautions are not taken.



CAUTION

indicates that minor personal injury can result if proper precautions are not taken.

NOTICE

indicates that property damage can result if proper precautions are not taken.

If more than one degree of danger is present, the warning notice representing the highest degree of danger will be used. A notice warning of injury to persons with a safety alert symbol may also include a warning relating to property damage.

Qualified Personnel

The product/system described in this documentation may be operated only by personnel qualified for the specific task in accordance with the relevant documentation, in particular its warning notices and safety instructions. Qualified personnel are those who, based on their training and experience, are capable of identifying risks and avoiding potential hazards when working with these products/systems.

Proper use of Siemens products

Note the following:



WARNING

Siemens products may only be used for the applications described in the catalog and in the relevant technical documentation. If products and components from other manufacturers are used, these must be recommended or approved by Siemens. Proper transport, storage, installation, assembly, commissioning, operation and maintenance are required to ensure that the products operate safely and without any problems. The permissible ambient conditions must be complied with. The information in the relevant documentation must be observed.

Trademarks

All names identified by ® are registered trademarks of Siemens AG. The remaining trademarks in this publication may be trademarks whose use by third parties for their own purposes could violate the rights of the owner.

Disclaimer of Liability

We have reviewed the contents of this publication to ensure consistency with the hardware and software described. Since variance cannot be precluded entirely, we cannot guarantee full consistency. However, the information in this publication is reviewed regularly and any necessary corrections are included in subsequent editions.

Security information

Siemens provides products and solutions with industrial security functions that support the secure operation of plants, systems, machines and networks.

In order to protect plants, systems, machines and networks against cyber threats, it is necessary to implement – and continuously maintain – a holistic, state-of-the-art industrial security concept. Siemens' products and solutions only form one element of such a concept.

Customer is responsible to prevent unauthorized access to its plants, systems, machines and networks. Systems, machines and components should only be connected to the enterprise network or the internet if and to the extent necessary and with appropriate security measures (e.g., use of firewalls and network segmentation) in place.

Additionally, Siemens' guidance on appropriate security measures should be taken into account. For more information about industrial security, please visit <https://www.siemens.com/industrialsecurity>.

Siemens' products and solutions undergo continuous development to make them more secure. Siemens strongly recommends to apply product updates as soon as available and to always use the latest product versions. Use of product versions that are no longer supported, and failure to apply latest updates may increase customer's exposure to cyber threats.

To stay informed about product updates, subscribe to the Siemens Industrial Security RSS Feed under <https://www.siemens.com/industrialsecurity>.

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1. Security Issues

(V1.2.0)

1.1. Personal Data Disclaimer

Siemens observes the principles of data protection, in particular the principle of data minimization (privacy by design). For this reason, the product only processes / stores technical functional data (e.g. time stamps) and no personal data. If the user links this data with other data (e.g. shift plans) or stores personal data on the same medium (e.g. hard disk) and thus establishes a personal reference, the user must ensure compliance with data protection regulations.

1.2. Security Concept

The physical access to the product (and related hardware) must be restricted (mounting inside a locked cabinet). Unauthorized people shouldn't get access to it or its connectors. The access to the web interface must be restricted to authorized people.

Both the overlayed systems (like PLC, HMI) and the underlaid systems (like IEDs, DP Slaves) must implement protections and interlocks to avoid damage of property or personal damage. These protections and interlocks must be done inside the single systems with the adequate configuration tool (like DIGSI, PDM).

Direct connection of the product to the internet must be avoided. In order to protect plants, systems, machines and networks against cyber threats, it is necessary to implement – and continuously maintain – a holistic, state-of-the-art industrial security concept. Siemens' products and solutions constitute one element of such a concept. For more information about industrial security, please visit the following web page:

<http://www.siemens.com/industrialsecurity>.

To increase the plant security, Siemens provides several Industrial Security Services. Please check the following web page to see which service is applicable for your project:

<https://new.siemens.com/global/en/products/services/industry/digital-industry-services/industrial-security-services.html>

NOTE

The product described here may use standard protocols or protocol versions that do not enable encryption or data integrity checking. This must be taken into account when designing the entire system.

NOTE

Please refer to "PCS 7 compendium f" and consider this with responsible and qualified personnel of your plant.

NOTE

The S7-Communication or other used protocols may have low protection, therefore it is recommended to protect the communication with a Scalance SC on the network.

1.3. Qualified Personnel

The product/system described in this documentation may only be operated by personnel qualified for the specific task in accordance with the relevant documentation, in particular its warning notices and safety instructions. Qualified personnel are those who, based on their training and experience, are capable of identifying risks and avoiding potential hazards when working with these products/systems.

1.4. Note of Usage and Misuse

Siemens products may only be used for the applications described in the catalogue and in the relevant technical documentation. If products and components from other manufacturers are used, these must be recommended or approved by Siemens. Proper transport, storage, installation, assembly, commissioning, operation and maintenance are required to ensure that the products operate safely and without any

problems. The permissible ambient conditions must be complied with. The information in the relevant documentation must be observed.

Please refer chapter “2 Safety Instructions” in manual “SIMATIC IPC227E or manual of superior product.

Connectivity products are not a replacement for a connected control system. It must be ensured that the connected system is fully functional and that experts for the system are available during operation.

1.5. Prevention of Misconfiguration

A misconfiguration of configured tags or the hardware properties can lead to malfunctions, including:

Product not reachable

Connected devices not reachable

Data not available or consistent

Sporadic data loss

Permanent data loss

To reduce this risk, a detailed signal loop test is highly recommended during commissioning.

NOTE

A CPU supports a specific number of network telegrams per second. This amount includes all telegrams. If the communication load is higher, it may cause delays or data loss. Please ensure that this limitation is considered.

1.6. Disclaimer

We have reviewed the contents of this publication to ensure consistency with the hardware and software described. Since variance cannot be precluded entirely, we cannot guarantee full consistency. However, the information in this publication is reviewed regularly and any necessary corrections are included in subsequent editions.

The product fulfils the functionality described in this document. Older releases of this document are invalid. Functionality which is not described or is explicitly marked as ‘not supported’ is not supported by this product or software release.

It is possible that superior products (like TIA Portal or PCS 7) provide a system standard functionality (Profibus or SFC). If this functionality isn’t handled inside this document, the system behavior of the product can differ from the superior system.

For more information about this topic, please contact:

<mailto:function.blocks.industry@siemens.com>.

1.7. System Hardening

To increase the security of the plant, we recommend a system hardening of this system and all other systems. Due to a large variety of needs/requirements, it is not possible to suggest a concrete hardening strategy.

For this device, please refer to the TIA Portal / PCS 7 manual and the hardware manual. Consider the strategy with responsible and qualified personnel of your plant. System hardening requires a deep knowledge about the guidelines, environment and needs on site. For this reason, system hardening isn’t covered by warranty or the support agreement. Please note a wrong hardening configuration can impact the functionality.

1.8. Integrity of Delivery

Please ensure the integrity of the delivered components when hardware is arrived and unpacked. To ensure the integrity, we seal the packaging of hardware components and the storage volumes with the following seal.

Figure 1-1: Valid Seals



Invalid seals can be identified by:

The corners are damaged from removing the seal

A removed seal leaves a pattern on the underground. If the seal is placed again, that pattern is missing in the background.

The seal was broken.

Figure 1-2: Invalid seals



NOTE

Please ensure that all components contain a valid seal. If the seals are missing or broken, do not connect the device or data volume. In this case, please contact us!

NOTE

For software components supplied, we strongly recommend validation to ensure data integrity before initial use. MD5 checksums for all product versions are available from us at function.blocks.industry@siemens.com

1.9. Updates

New product versions are published in SIOS in delivery releases. A e-mail notification should therefore be set up within SIOS for the corresponding product article number. This means that you automatically receive a notification when a successor version of the product is available. Alternatively, updates should be requested on our hotline on a regular base. Depending on the update cause (feature extensions, bug fixing, closing security issues, ...), the installation of the update is strongly recommended.

As software updates (e.g. new features, new compatibilities or closing security vulnerabilities) are based on the latest product version, only the latest product version should be used.

Some products require a Support Agreement for getting free of charge updates. If a valid Support Agreement exists, the update can be requested on our hotline.

NOTE

The availability of security updates should be checked by e-mail notification or permanently manually to ensure proper and secure operation.

NOTE

When using or updating PCS 7 or PCS 7 related products, make sure to refer to the corresponding PCS 7 documentation, such as the PCS 7 compendium.

NOTE

Check in SIOS or ask support whether the product described here supports SUS.

1.10. Cybersecurity information

Siemens provides products and solutions with industrial cybersecurity functions that support the secure operation of plants, systems, machines and networks.

In order to protect plants, systems, machines and networks against cyber threats, it is necessary to implement – and continuously maintain – a holistic, state-of-the-art industrial cybersecurity concept. Siemens' products and solutions constitute one element of such a concept.

Customers are responsible for preventing unauthorized access to their plants, systems, machines and networks. Such systems, machines and components should only be connected to an enterprise network or the internet if and to the extent such a connection is necessary and only when appropriate security measures (e.g. firewalls and/or network segmentation) are in place.

For additional information on industrial cybersecurity measures that may be implemented, please visit www.siemens.com/cybersecurity-industry.

Siemens' products and solutions undergo continuous development to make them more secure. Siemens strongly recommends that product updates are applied as soon as they are available and that the latest product versions are used. Use of product versions that are no longer supported, and failure to apply the latest updates may increase customer's exposure to cyber threats.

To stay informed about product updates, subscribe to the Siemens Industrial Cybersecurity RSS Feed under <https://www.siemens.com/cert>.

1.11. Third party components

This product may contain 3rd party components. The included open-source components can be found in the delivery state (delivery USB stick or archive for online delivery) including the source code and any readme files.

The product also contains the following commercial 3rd party products:

Vendor	Component	Version
Under construction	Under construction	Under construction

1.12. Recycling and disposal

The products are low in pollutants and can be recycled. For environmentally compliant recycling and disposal of your electronic waste, please contact a company certified for the disposal of electronic waste.

The following must be observed when removing and recommissioning the product:

- The system must be in safe condition. Backups must be created
- Removal of the product components from the superordinate SIMATIC project
- In the case of a hardware product, the device must be switched off in accordance with the applicable safety regulations. The cables can then be removed and the device can be dismantled of the rack
- In the case of a software product, the software must be uninstalled
 - Software with installer (setup) is removed by calling up the installer

- Software without installer (setup), requires manual removal of the manually installed components

1.13. Open-Source Components

Notes to Open-Source components of this product: Portions include Open-Source Software.

See ReadMe_OSS.txt for details. For further details, see the following pattern as agreed with A&D GC in 2007. Please consider the ReadMe_OSS.txt as part of this manual.

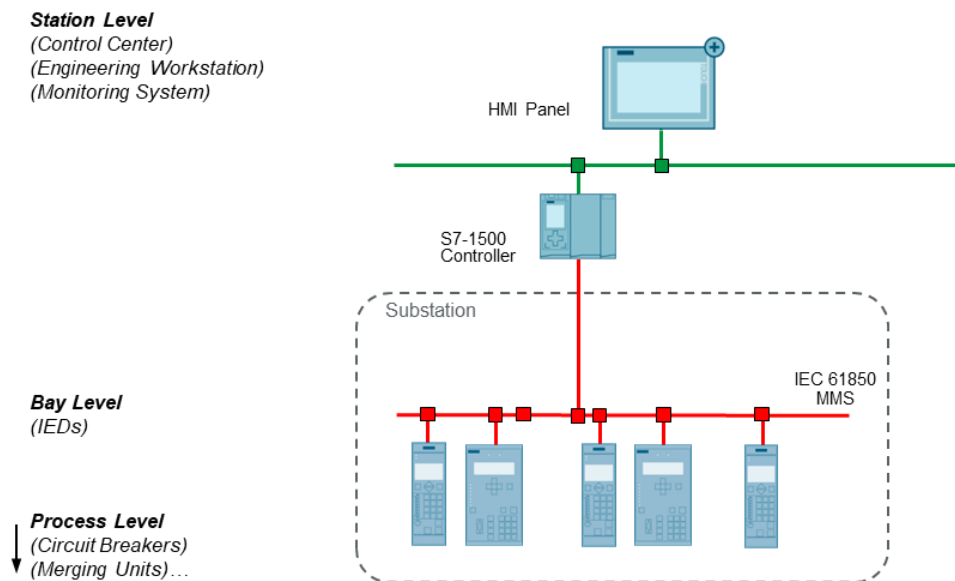
2. System Requirements and Use Conditions

2.1. Overview

This document is valid for IEC 61850 Client Library – TIA Portal only. The IEC 61850 Client Library connects IEC 61850 devices from different vendors to the PLC.

The gateway uses S7 1500 CPU or CP Ethernet RJ45 interfaces: The used interface is connected to the Ethernet network with the protection relays using the protocol IEC 61850 MMS (Manufacturing Messaging Specification).

The IEC 61850 Client Library supports the communication protocol IEC 61850 MMS Edition 1 and Edition 2.1.



2.2. Software

TIA Portal V19

IEC 61850 device engineering tool

IEC Browser

Most parts of this manual are also valid for TIA V13, V13.1, V14, V15, V15.1, V16, V17, V18. For more information, please contact function.blocks.industry@siemens.com.

2.3. Library

The mainline of this library is for the usage with CPU S7 1500. Depending on the project environment, other versions can be requested.

For more information, please contact function.blocks.industry@siemens.com.

2.4. Hardware

The following Hardware is supported with the IEC 61850 block library.

- S7-1500 - Advanced Controller (e.g.: 1511, 1513, 1515, 1516, 1517, 1518)
- S7-1500 - Software Controller (e.g.: 1505SP, 1507S, 1508S)

- S7-1500 - Distributed Controller (e.g.: 1510SP, 1512SP)
- S7-1500 - Redundant Controller (e.g.: 1513R, 1515R, 1517H)

Application examples for S7-1511 and S7-1517H are in scope of delivery. Configuration at other controllers might differ from the example projects.

2.5. Maximum number of instances per S7-1500 CPU

Because the S7 CPU handles the IEC 61850 protocol, the IEC61850_COM/ IEC61850_RCOM block consumes a lot of scan time and memory. (The CP module only handles the deeper Layers ISO on TCP and TCP.) Therefore, the IEC 61850 library is limited to up to 8 connections per CPU.

However, the maximum possible connections are also limited by the CPU type.

2.6. General performance restrictions

The maximum number of IEC 61850 tags, which can be handled by the library, depends on the used CPU and the number of connected IEC 61850 server devices.

The TIA Station Gateway offers a highly performant and flexible option to connect up to 50 IEC 61850 servers. This might be a suitable alternative. For more information see chapter 7.5.1.

For further information, please contact function.blocks.industry@siemens.com

2.7. Redundancy limitation

If the library is used in a redundant system, the cases in which a redundancy switch over or a return to redundancy occur should be considered.

- In case of a redundancy switch over, the IEC61850_RCOM block reconnects automatically. The reconnection duration depends on function block settings and the cyclic interval but will be at least 10 seconds.
- Since the IEC 61850 connection technically behaves like a standard S7 connection, the connection is reinitialized after a redundancy return and should only be performed with the necessary safety precautions.

-

NOTE

In order to use the blocks, a license for each plant/project is required.

Additional functionality can be implemented on request.

2.8. Whats new or different?

Since version 18 (Lib_TIA_V18_IEC61850_V2 or Lib_TIA_V19_IEC61850) of the IEC 61850 Client Library the IEC 61850 Edition 2.1 is supported. This is also backwards compatible to Edition 2.0 and Edition 1.

With Edition 2.1 the maximum string lengths for the IEC 61850 addresses were increased. This affects the IED Name and signal paths. The following table lists the maximum string lengths.

Table 2-1 Maximum string lengths

String type	<V18	>=V18
IED Name	24	64
Signal path	52	128
Complete (IED Name + signal path)	64	128

2.9. Support Agreement

A 5-year support agreement (MLFB: 9LA11106PC102CA0) is mandatory for this library. This support agreement covers questions about general topics, system handling and troubleshooting via phone or email. This support agreement provides library updates for new versions of TIA Portal. The license of the library and the support agreement are bound to one plant.

Without a valid support agreement, no free of charge support or updates are possible. If support via email or phone is required, an extended support package has to be ordered if the service contract is run out.

A template of the support agreement document can be found in the appendix.

3. Installation

3.1. Requirements

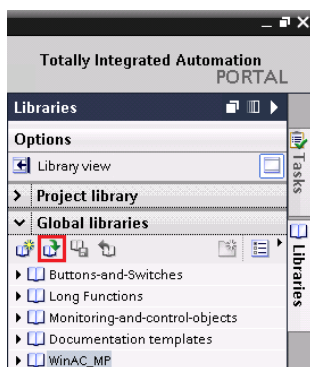
This version of IEC 61850 Client Library requires TIA V18 or V19.

3.2. TIA Portal – Open global library

The IEC 61850 Client library is a global TIA Portal library with the file name Lib_TIA_v18_IEC61850_Client_1500_V2.zal18 or Lib_TIA_v19_IEC61850_Client_1500.zal19.

The library has to be opened in the project view of TIA Portal. Open the library with the option “Open as read-only” in the open file dialog.

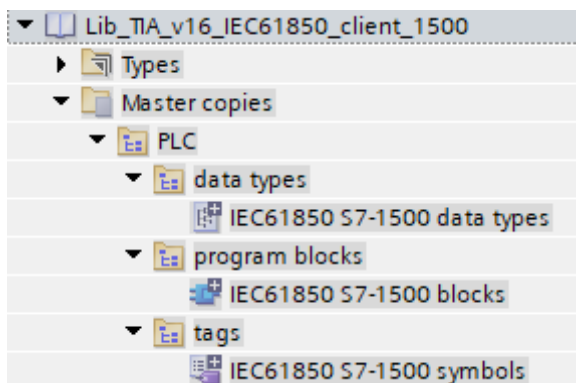
Figure 3-1 Open library



After the library was opened, you can find all objects for using the library in the directory “Master copies”. This directory contains:

- The IEC 61850 data types
- The IEC 61850 symbols
- The IEC 61850 program blocks

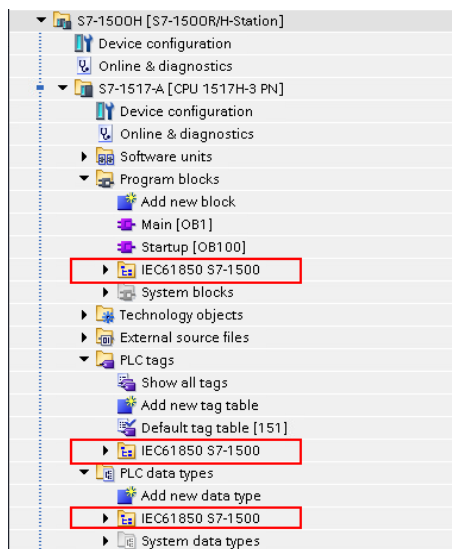
Figure 3-2 Library structure



Copy the PLC elements from the library to your PLC object in the project which should use the IEC 61850 communication.

- IEC 61850 symbols → PLC tags
- IEC 61850 data types → PLC data types
- IEC 61850 program blocks → Program blocks

Figure 3-3 PLC



Compile all your program blocks (Software / Rebuild all blocks).

After the compile action is finished, the following system blocks should be added to your PLC.

Table 3-1 System blocks

BLOCK NO	NAME	PLC-Type
FB 700	Program_Alarm	S7-1500
FB 1030	TSEND_C	S7-1500
FB 1031	TRCV_C	S7-1500
FC 108	RH_GetPrimaryID	S7-1500 R/H

3.3. Hardware Identifier

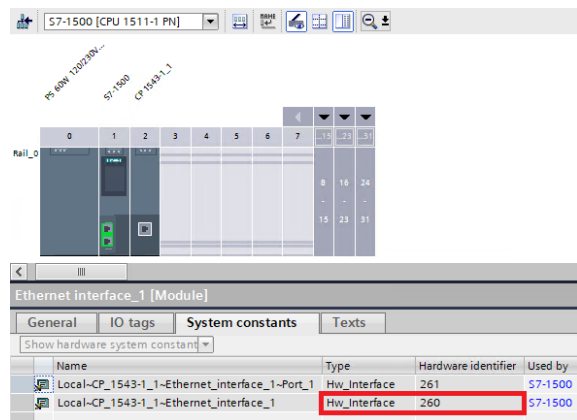
Like mentioned in the introduction, the IEC 61850 client library can be used with various S7-1500 PLCs. As there are different PLC types available, depending on the product used, some points must be considered. Even before programming starts, the network interface to be used must be defined. As not every ethernet interface is suitable for IEC 61850 communication, the next chapter contains information about which ports can be used and how to identify the required hardware identification IDs.

3.3.1. S7-1500 Advanced Controller

In case of an S7-1500 Advanced Controller (e.g., S7-1511-1 PN), the internal PN-Interface as well as the ethernet interface of an external CP can be used for IEC 61850 communication. To get the correct hardware identifier, please select the desired port and switch to "System constants". The hardware identifier of the

superior ethernet interface must be used for IEC communication, so make a note of this number for programming which is explained in chapter 3.4.

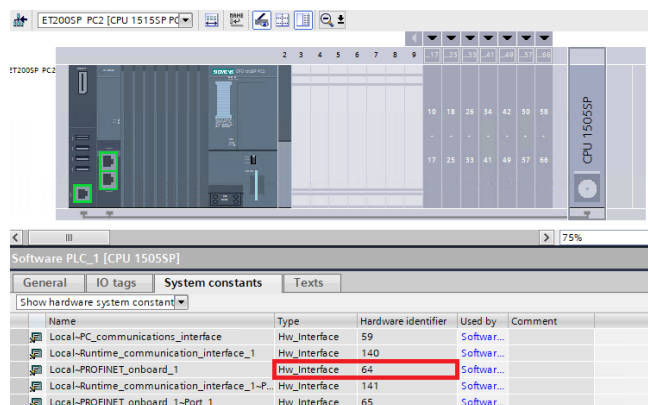
Figure 3-4



3.3.2. S7-1500 Software Controller

In case of an S7-1500 Software Controller (e.g., S7-1505SP), the internal PN-Interface can be used for IEC 61850 communication. The ethernet interface, which is used for the Windows environment, cannot be used for IEC 61850 communication with this library. To get the correct hardware identifier, please select the desired port and switch to "System constants". The hardware identifier of the superior ethernet interface must be used for IEC communication, so make a note of this number for programming which is explained in chapter 3.4

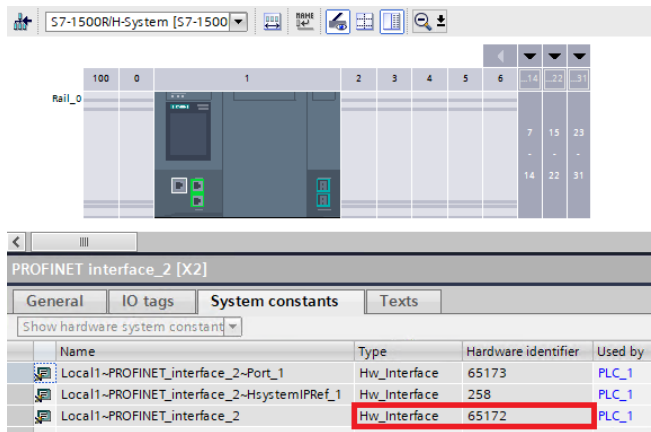
Figure 3-5



3.3.3. S7-1500 R/H Controller

In case of an S7-1500 R/H Controller (e.g., S7-1517H-3 PN) the internal PN-Interfaces can be used for IEC 61850 communication. To get the correct hardware identifier, please select the desired port and switch to "System constants". The hardware identifier of the superior ethernet interface must be used for IEC communication, so make a note of these numbers at **both** PLCs for programming which is explained in chapter 3.4.

Figure 3-6



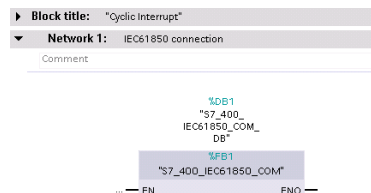
3.4. TIA Portal – Engineering

An extensive function- and performance test was performed with up to eight IEDs. The performance (reaction time and manageable number of signals) depends on the CPU performance and engineering.

3.4.1. Setup IEC 61850 communication

- Add a new cyclic organization block (e.g., OB35) to your PLC
- Add a new start up organization block (e.g., OB100) to your PLC
- Add a new Function Block (FB) to your PLC.
- Open the organization blocks in editor, drag the new FB into a network and add the instance data block to your program.

Figure 3-7 Cyclic OB – Call FB



- Open the function block in the editor.
- Select the function block "IEC61850_COM" underneath the IEC 61850 group directory in your PLC and drag it to a network in your function block. One block is required for each IEDs.
- Add the instance data block to your program. The IEC61850_COM block's inputs and outputs are explained in detail in chapter 5.1.
- Configure the block inputs for communication with the IED:

IP_ADDR

- Set the IED IP-Address of the IED at the input IP_ADDR.

CON_ID

Specify a unique connection ID at the input CON_ID for every IEC61850_COM block, starting at "1".

Interface_ID

- Enter the Interface_ID identified in chapter 3.3.

MODE:

Here you can choose the mode for the cyclically reading of tags. It is important to choose the correct mode for your application.

- Mode "1" – DATASET
- Mode "2" – LIST

For initial test purposes the LIST Mode should be used, as it is the easiest way to set up an IEC 61850 connection. However, if your device supports DataSets, we recommend using DATASET mode for productive operation, because it is the fastest way for communication.

NOTE

For more information see block description at chapter 5.1 and application example 4.2.

Reporting

If your IEC 61850 device supports Reporting, enter the DataSet address for reporting, containing the list of Report values, in the input **RP_DSADDR**. (e.g., "CTRL/LLN0\$\$DataSet2"). Only the complete address is allowed.

NOTE

For more information see block description at chapter 5.1 and application example 4.2.

IED_NAME

Configure the Input **IED_NAME** with the IEC 61850 device name. (Please note, all addresses and names are case sensitive).

SAMPLE_T

Set the cycle time of the OB which calls the communication block to **SAMPLE_T**. This input should never be 0.0. It's used for TIMOUT calculation. (e.g., 100 ms Cycle time means **SAMPLE_T** = 0.1).

EN_MSG

To enable WinCC messaging, set **EN_MSG** to 1.

STRT_COM

Set the **STRT_COM** input to "OFF" until the engineering work is done; to start communication, set it to "ON".

Please follow up by adding at least one **RD_XXX** block. This procedure is explained in chapter 3.4.2..

3.4.2. Configure Tag blocks

The IEC 61850 Client library is designed for a very flexible use. On one hand, there is the **IEC61850_COM** block which handles the IEC 61850 protocol block and management functionality. On the other hand, there are several blocks which can be connected to the **IEC61850_COM** block, called tag blocks. Each of the tag blocks represents one value in a special data format (e.g., INT, REAL, BOOL).

NOTE

At least one **IEC_RD_XXX** block has to be connected to the **IEC61850_COM** block.

Currently, the library supports the following tag blocks which read IEC 61850 tags cyclically:

- IEC_RD_BO (BOOL)
- IEC_ALARM (BOOL with timestamp)
- IEC_RD_BS (BITSTRING)
- IEC_RD_IN (INT)
- IEC_RD_RL (REAL)
-
-
- The following tag blocks read values on change with the IEC 61850 reporting mechanism
- IEC_REPORT (BOOL on change with timestamp)

Switches can be controlled with:

- IEC_CTRL (uses IEC 61850 control mechanism)

Additionally, tags can be written (e.g., substitution values) with the following blocks

- IEC_WR_BO (BOOL)
- IEC_WR_BS (BITSTRING)
- IEC_WR_IN (INT)
- IEC_WR_RL (REAL)

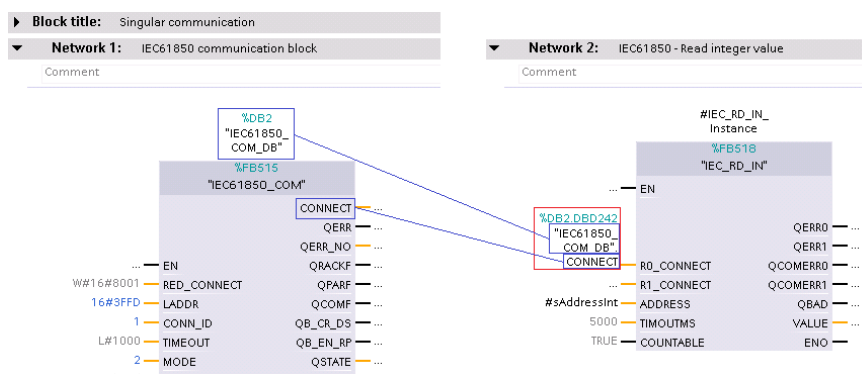
3.4.3. Engineer Tag blocks

Add e.g., one IEC_RD_IN block. Set R0_CONNECT to the data block value IEC61850_COM_DB.CONNECT.

NOTE

For the following tag blocks always use the same order of R0_CONNECT.

Figure 3-8 Set R0_Connect to DB value "CONNECT"



Enter the IEC 61850 address string on input "ADDRESS".

NOTE

Please note you have to replace "\$" with "\$\$".

NOTE

These IEC 61850 address strings can also be generated and copied via the IEC Browser configuration tool. For further information see chapter 3.5.

Simple tag blocks

- IEC_RD_BS (BITSTRING),
- IEC_RD_IN (INT),
- IEC_RD_RL (REAL)
- IEC_RD_BO (BOOL)

The full path including the object attribute has to be entered, starting with the IEC 61850 domain name.

E.g., 'CTRL/LLN0\$\$ST\$\$Loc\$\$stVal' (for a BOOL tag block).

Complex tag blocks

- IEC_ALARM (BOOL with timestamp),
- IEC_REPORT (BOOL on change with timestamp)
- IEC_CTRL (uses IEC 61850 control mechanism)
-

NOTE

Only the path to the object name itself has to be entered, e.g., 'CTRL/LLN0\$\$ST\$\$Loc' (for a REPORT tag block).

NOTE

In Case of IEC_CTRL, the path excluding Oper/SBO object has to be set up: e.g., 'CTRL/LLN0\$\$CO\$\$LEDs\$'.

Interface				
	Name	Data type	Offset	Default value
1	▶ Input			
2	▼ Output			
3	QValueBool	Bool	...	false
4	QValueInt	DInt	...	0
5	QValueReal	Real	...	0.0
6	QValueBS	DWord	...	16#0
7	<Add new>			
8	▶ InOut			
9	▼ Static			
10	sComledName	String[24]	...	'IED_0002'
11	sEnMsg	Bool	...	true
12	IEC_RD_IN_Instance	"IEC_RD_IN"	...	
13	IEC_RD_BO_Instance	"IEC_RD_BO"	...	
14	IEC_RD_RL_Instance	"IEC_RD_RL"	...	
15	IEC_RD_BS_Instance	"IEC_RD_BS"	...	
16	sAddressBool	String[52]	...	'CTRL/LLN0\$\$ST\$\$LEDs\$\$stVal'
17	sAddressInt	String[52]	...	'CTRL/LLN0\$\$ST\$\$LEDs\$\$origin\$\$orCat'
18	sAddressReal	String[52]	...	'MEAS/MMTR1\$\$CF\$\$SupWh\$\$pulsQty'
19	sAddressBS	String[52]	...	'CTRL/LLN0\$\$ST\$\$LEDs\$\$q'

Figure 3-9 Example: Tag block configuration

RP_CTRL_ADDR2

In case of DATASET mode and reporting mechanism, an additional report control block is required. Set an IEC 61850 path to report control block at the input RP_CTRL_ADDR2, which is different than the RP_CTRL_ADDR1.

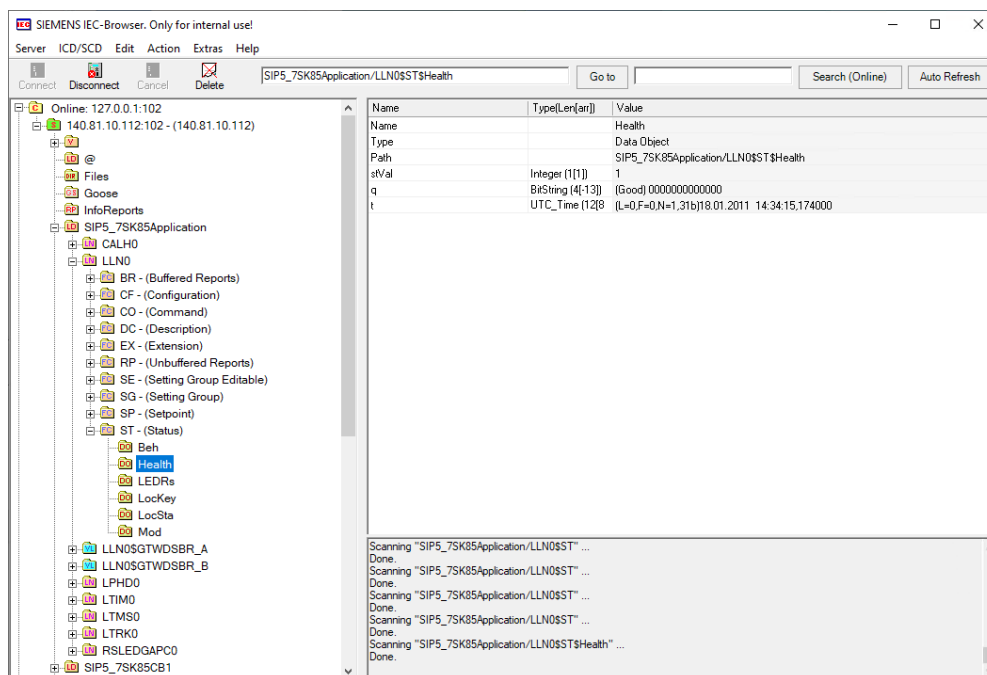
3.5. Configuring IEC 61850 addresses

With the „IEC Browser“ it is possible to access the data structure on an IEC 61850 server. It is possible to connect to an online device and read all the data objects, and even change some data objects.

Alternatively, an ICD or SCD can be imported to show the data structure of one or multiple devices. Here the available information is limited.

Furthermore, it is possible to configure datasets which can be read from the IEC 61850 Client Library with the reporting mechanism.

Figure 3-1 IEC Browser

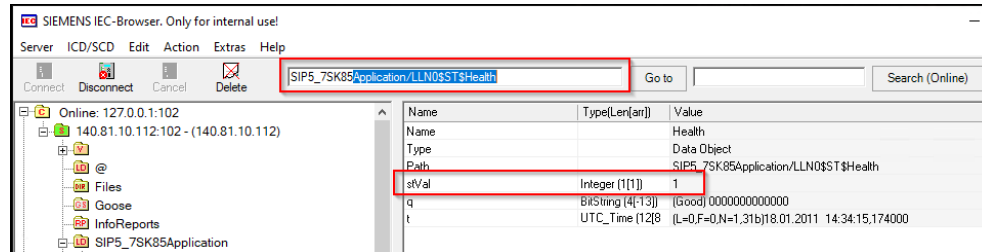
**NOTE**

The "IEC Browser" can be downloaded from the Siemens Industry Online Support (SIOS).

After you established a connection to an IED it is possible to copy and paste the IEC61850 addresses from the "IEC Browser". For this select the data object you want to configure in the IEC 61850 Client Library and

copy the path from the textbox at the top. In this example it is the current health status of the device "Application/LLN0\$ST\$Health". Furthermore, you have to append the relevant signal "stVal" and the additional character "\$" has to be replaced with "\$\$". It is not necessary to copy the IED Name, in this example "SIP5_7SK85".

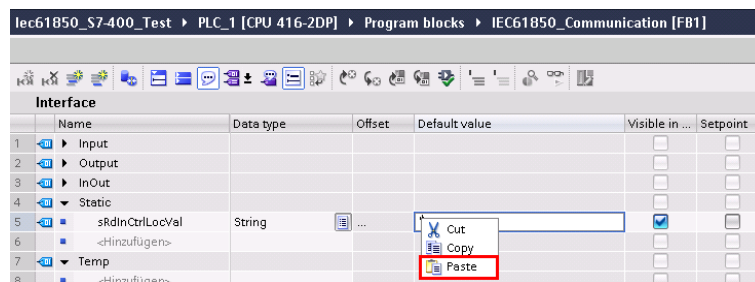
Figure 3-2 Copy data object address from IEC Browser



So the complete address for the signal to copy to the TIA portal program block for this example is "Application/LLN0\$\$ST\$\$Health\$\$stVal".

After this, switch to your TIA Portal program block and paste the address value.

Figure 3-3 Paste address to TIA Portal program block



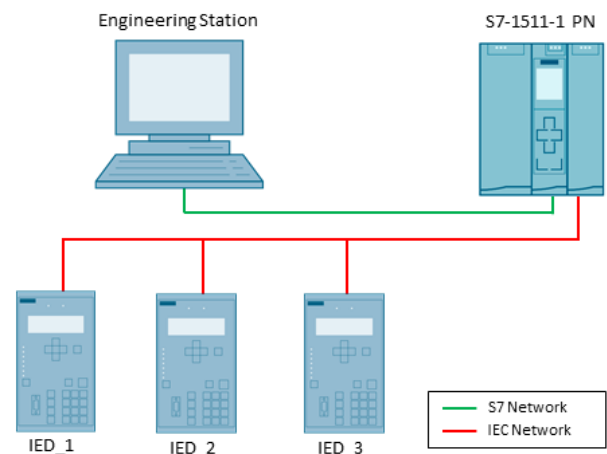
4. Application Examples

In the following two examples, different possible configuration options are shown and explained step by step. The example projects are part of the scope of delivery and should show possibilities how the library can be used correctly.

4.1. S7-1500 Advanced PLC

The first example shows the engineering basics of the IEC 61850 client library used to connect three identical SIPROTEC4 protection relays to a S7-1500 Advanced controller with a CP-Modul.

Figure 4-1 Example 1 - Architecture

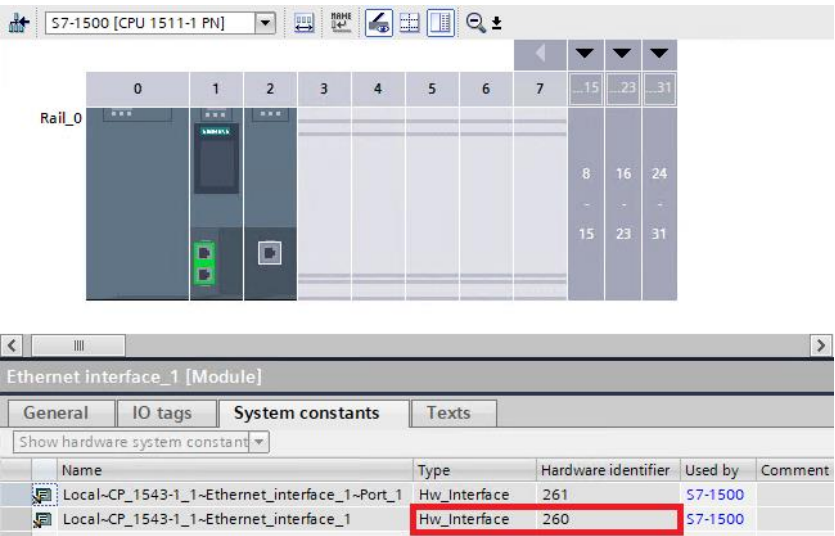


Because we want to keep the configuration as simple as possible, we will perform a small configuration with a couple of common values including BOOL, FLOAT, INTEGER which are exchanged cyclical, every 400 ms with the IED. Furthermore, we will engineer a control function as it is usually used for switches.

4.1.1. Hardware configuration

The first step is to configure the hardware configuration for the used PLC as usual in TIA Portal – in this example an S7-1511 advanced controller with an external CP. As we want to use the CPs ethernet port for IEC 61850 communication, we need to note the corresponding Hw_Interface ID. In this case it is “260” as marked in the following picture.

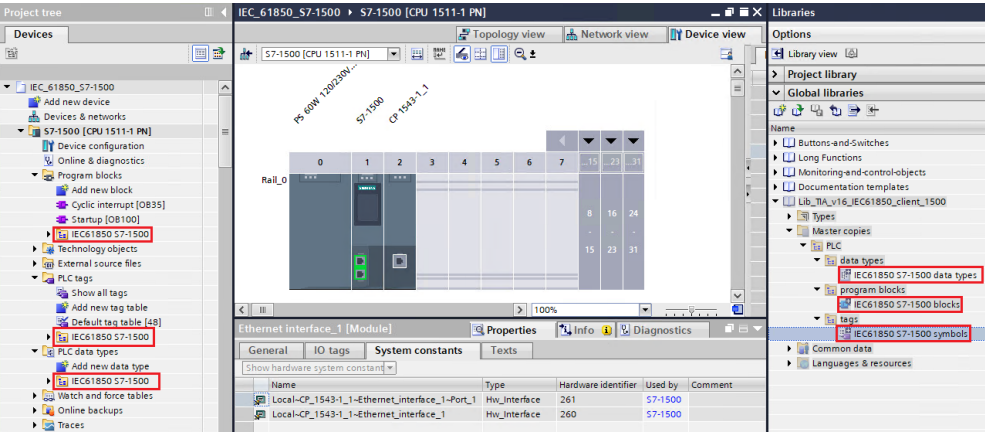
Figure 4-2 HW Interface ID / Singular PLC



4.1.2. Engineering

After the hardware configuration is done, drag & drop the master copies of the IEC 61850 client library to the corresponding folders at the PLC.

Figure 4-3 Master copies S7-1500



Create a startup OB (e.g., OB100) and a cyclic OB (e.g., OB35, 100 ms) which calls a FB named "IED_0001". For one complete data exchange cycle, four PLC cycles are needed. In the FB "IED_0001" one IEC61850_COM block must be inserted. Before we can start the configuration of the target values, the general connection properties need to be set correctly as described in chapter 3.4.1.

The following Inputs must be set in this example:

Table 4-1

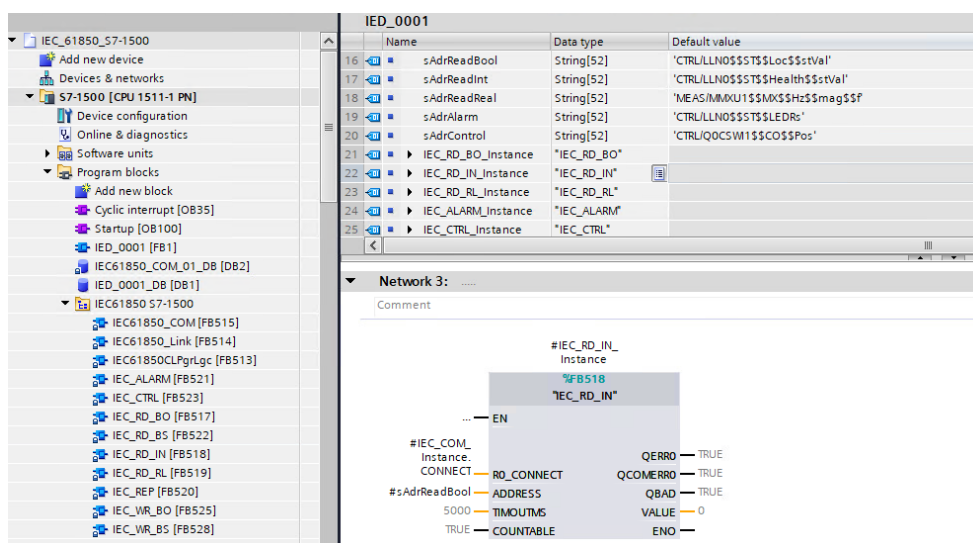
Input	Value	Comment
IP_ADDR	140.80.161.101	IP Address of IED
CON_ID	1	1 st Connection
Interface_ID	260	As identified before
MODE	2	Only cyclic data exchange
IED_NAME	IED_0001	IED name from SCD/ICD File
DS_CONF_OP	1	Dynamic Dataset supported
DS_NAME	CTRL/LLN0\$\$DatasetIECLib	Name for Dyn-Dataset
SAMPLE_TIME	0.1	Cycle time in seconds
EN_MSG	0	No alarming
START_COM	Bool PLC Tag "gvStartCom"	To start communication
RESET	Bool PLC Tag "gvResetCom"	To reset communication

Figure 4-4 IEC61850_COM configuration

- Boolean
- Integer
- Float
- Bool with timestamp
- Control-Structure

Create static string variables for each corresponding IEC path of the target values and instance dbs for the tag blocks as shown in picture. At this point it is important to consider the right IEC path for each block type. While all IEC_RD_XX-blocks need total IEC paths e.g., 'CTRL/LLN0\$\$ST\$\$Loc\$\$stVal' the IEC_ALARM and the IEC_CONTROL block needs the whole structure path, e.g., 'CTRL/Q0CSW1\$\$CO\$\$Pos' without absolute target value.

Figure 4-5 IEC tag configuration



The next step is to connect the RO_CONNECT input to the CONNECT output of the IEC61850_COM block and the ADDRESS input with the matching static string containing the IEC 61850 path.

Finally, we need to create two more PLC tags to use the IEC_CTRL block. Connect one Boolean PLC tag named "ctrlValue" with the input CTRLVAL and another Boolean PLC tag named "ctrlTrigger" to the CTRL input.

At last, copy the IED_0001 block twice and rename it for the other two IEDs. If all IEDs are the same, just the IP address, IED name and the CON_ID input must be modified individually.

When the engineering is completed, the program can be loaded to the PLC for testing purposes. The IEC 61850 communication can be started by modifying the START_COM input of the IEC61850_COM block.

After a few seconds, the QSTATE output is expected to show "12", which means cyclic communication is in progress – the engineered tag blogs should now show the correct values of the IEDs – it's recommended to compare the online values with a suitable tool, e.g., "IEC Browser".

4.1.3. Conclusion

Congratulations, the first IEC 61850 communication to the S7-1500 PLC is finished. Even if the communication is running fine, you should consider some points before putting the PLC to productive use.

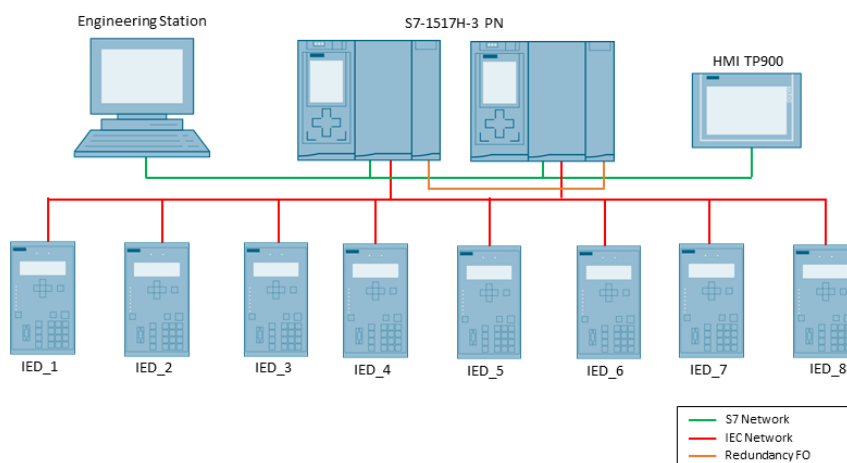
As the cyclic IEC 61850 communication leads to a high consumption of resources, this kind of communication is only suitable for a small amount of IEDs depending of the required update interval, the number of tags, the used PLC model and of course the other task of the PLC.

As a more powerful PLC is not always a possible solution, you should take the second application sample into account, where the "Report" mechanism for data exchange of the IEC 61850 protocol is being used. This has big advantages regarding the performance compared to the cyclic data exchange (also called "Polling").

4.2. S7-1500H Redundant PLC

The second example shows a more complex automation system including a redundant S7-1500 PLC, several connected IEDs and an HMI panel. To reduce the communication load for the PLC to a minimum, report mechanism is being used for data exchange.

Figure 4-6 Example 2 - Architecture

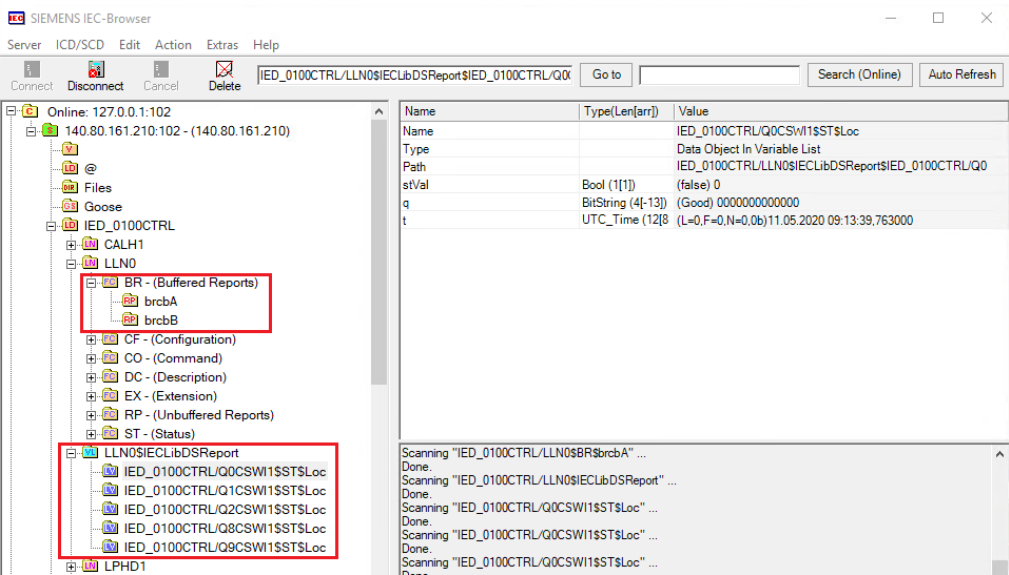


4.2.1. Preconditions

To use the report mechanism for data exchange, it is required that the IEDs have a preconfigured static dataset which contains all tags that shall be reported to the client. Additionally, two buffered report control

blocks are necessary as shown in Figure 4-6. Usually, a separate engineering tool is required to configure the IED, e.g., for SIPROTEC devices you need to use the DIGSI software.

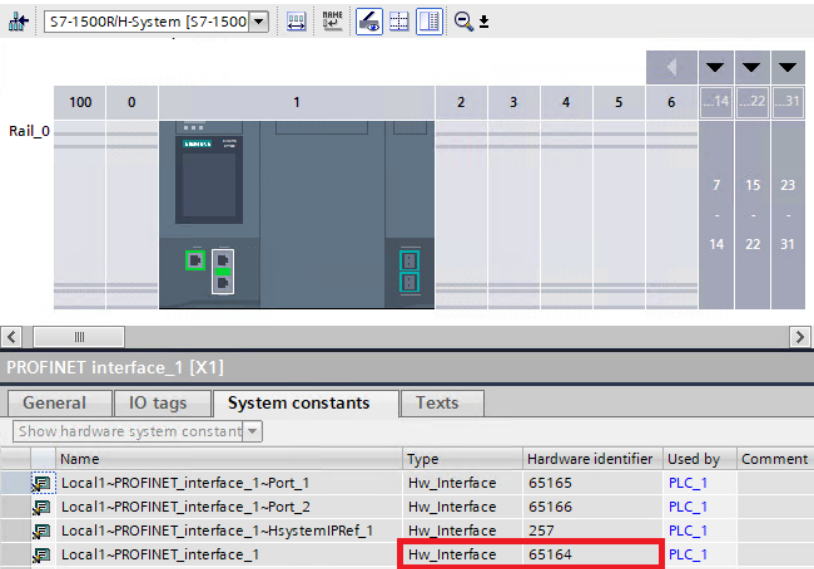
Figure 4-7 Static DataSets & BRCBs



4.2.2. Hardware configuration

The first step is to configure the hardware configuration for the used PLC as usual in the TIA Portal – in this example an S7-1517H redundant controller. As we want to use both of the PLCs for IEC 61850 communication, we need to note the corresponding Hw_Interface IDs. In this case, it is “65164” as marked in the following picture for the first PLC and “65364” for the second PLC. Configure the HMI panel for alarming of the S7-1517H PLCs.

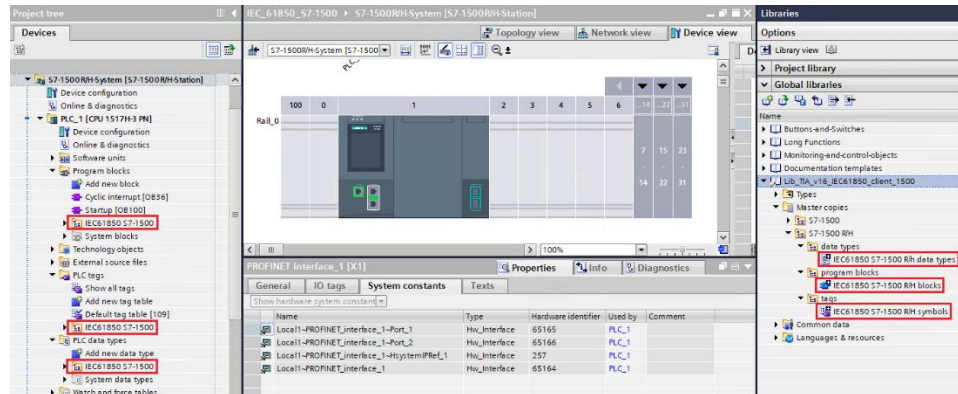
Figure 4-8 HW Interface ID / Redundant PLC



4.2.3. Engineering

After the hardware configuration is done, drag & drop the master copies of the IEC 61850 client library to the corresponding folders at the PLC.

Figure 4-9 Master copies S7-1500 R/H



Create a startup OB (e.g., OB100) and a cyclic OB (e.g., OB36, 50 ms), which calls a FB named "IED_0100". For one complete data exchange cycle, four PLC cycles are needed. In the FB "IED_0100" one IEC61850_RCOM block must be inserted. Before we can start the configuration of the target values, the general connection properties need to be set correctly as described in chapter 3.4.4.

The following inputs must be set in this example:

Table 4-2

Input	Value	Comment
IP_ADDR	140.80.161.211	IP Address of IED
CON_ID	1	1 st Connection
INTERFACE_ID1	65164	As identified before
INTERFACE_ID2	65364	As identified before
RECON_TIMEOUT	T#15s	Reconnect delay 15s
MODE	1	Dataset mode
IED_NAME	IED_0101	IED name from SCD/ICD File
DS_CONF_OP	1	Dynamic Dataset supported
DS_NAME	CTRL/LLNO\$\$DatasetIECLib	Name for Dyn-Dataset
RP_CONF_OP	1	
RP_DSADDR	CTRL/LLNO\$\$DatasetReport	
RP_CTRL_ADDR1	CTRL/LLNO\$\$BR\$\$brcbA01	
RP_CTRL_ADDR2	CTRL/LLNO\$\$BR\$\$brcbB01	
SAMPLE_TIME	0.05	Cycle time in seconds
EN_MSG	1	Alarming enabled
START_COM	Bool PLC Tag "gvStartCom"	To start communication
RESET	Bool PLC Tag "gvResetCom"	To reset communication

After the configuration of the IEC61850_RCOM block is completed, the required tag blocks for the IED can be engineered. At least one "Read"-Tag is mandatory. In this sample application, several sample values are engineered to show the engineering principles:

- Boolean

- Integer
- Float
- 5x Boolean Report

Create static string variables for each corresponding IEC path of the target values, and instance dbs for the tag blocks as shown in Figure 4-10. At this point, it is important to consider the right IEC path for each block type. While all IEC_RD_XX-blocks need total IEC paths e.g., 'CTRL/LLN0\$\$ST\$\$Loc\$\$stVal' the IEC_REP block needs the whole structure path, e.g., 'CTRL/QOCSWI1\$\$ST\$\$Loc' without an absolute target value.

Figure 4-10 IEC Report configuration

The screenshot displays the 'HW Config' window for a SIMATIC 300 station. The left pane shows the project tree with 'PLC_1 [CPU 1517H-3 PN]' selected. The right pane shows the 'IED_0101' configuration table and a ladder logic diagram.

Name	Data type	Default value	Retain	Accessible f...	Writa...	Visible in ...
sAdReadBool	String[52]	'CTRL/LLN0\$\$ST\$\$Loc\$\$stVal'	Non-retain	☒	☒	☒
sAdReadInt	String[52]	'CTRL/LLN0\$\$ST\$\$Health\$\$stVal'	Non-retain	☒	☒	☒
sAdReadReal	String[52]	'MEAS/IMMU1\$\$MMSHt\$\$smag\$...	Non-retain	☒	☒	☒
sAdReport5	String[52]	'CTRL/Q9CSW1\$\$ST\$\$Loc'	Non-retain	☒	☒	☒
sAdReport4	String[52]	'CTRL/Q8CSW1\$\$ST\$\$Loc'	Non-retain	☒	☒	☒
sAdReport3	String[52]	'CTRL/Q2CSW1\$\$ST\$\$Loc'	Non-retain	☒	☒	☒
sAdReport2	String[52]	'CTRL/Q1CSW1\$\$ST\$\$Loc'	Non-retain	☒	☒	☒
sAdReport1	String[52]	'CTRL/QOCSW1\$\$ST\$\$Loc'	Non-retain	☒	☒	☒
IEC_RCOM_Instance	'IEC61850_RCOM'			☒	☒	☒
IEC_RD_BO_Instance	'IEC_RD_BO'			☒	☒	☒
IEC_RD_RL_Instance	'IEC_RD_RL'			☒	☒	☒
IEC_RD_IN_Instance	'IEC_RD_IN'			☒	☒	☒
IEC_REP_Instance5	'IEC_REP'			☒	☒	☒
IEC_REP_Instance4	'IEC_REP'			☒	☒	☒
IEC_REP_Instance3	'IEC_REP'			☒	☒	☒
IEC_REP_Instance2	'IEC_REP'			☒	☒	☒
IEC_REP_Instance1	'IEC_REP'			☒	☒	☒

The ladder logic diagram shows the configuration of the IEC61850_COM block. It includes inputs for EN, R0_CONNECT, ADDRESS, S000, TRUE, #sEnMsg, and false. It also shows outputs for QERRRO, QCOMERRRO, QBAD, ALARM, BS, TIMEST, MSG_DONE, QMSG_ERR, QMSG_SUP, MSG_STAT, MSG_ACK, and ENO. The diagram is labeled with '#IEC61850_COM [FB515]' and '#IEC61850_RCOM [FB516]'.

The next step is to connect the R0_CONNECT input to the CONNECT output of the IEC61850_COM block, and the ADDRESS input with the matching static string containing the IEC 61850 path.

At last, copy the IED_0101 block seven times and rename it for the other IEDs. If all IEDs are the same, just the IP address, IED name and the CON_ID input must be modified individually.

When the engineering is completed, the program can be loaded to the PLC for testing purposes. The IEC 61850 communication can be started by modifying the START_COM input of the IEC61850_COM block.

After a few seconds, the QSTATE output is expected to show "12", which means cyclic communication is in progress – the engineered tag blogs should now show the correct values of the IEDs and the HMI will show the alarms in case of a signal change at the report blocks – it's recommended to compare the online values with a suitable tool, e.g., "IEC Browser".

4.2.4. Conclusion

In this application example, the principles of using the IEC 61850 Client Library with reporting mechanism was explained, which is highly recommended to keep the PLC load on a reasonable level.

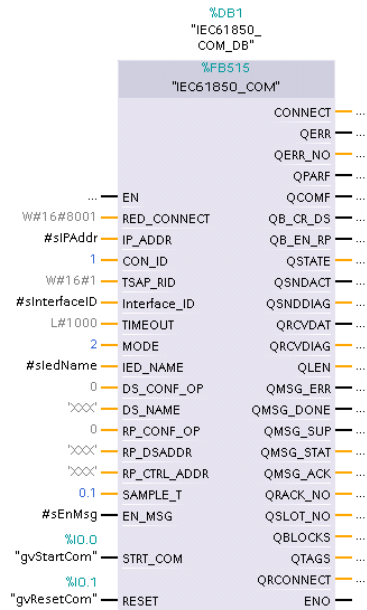
5. Block description

5.1. IEC61850_COM

5.1.1. Block screenshot

Figure 5-1 IEC61850_COM block S7-1500

7	Static		
8	sIedName	String[24]	'IED_0002'
9	sEnMsg	Bool	true
10	sEnTCON	Bool	true
11	sIPAddr	String[15]	'140.80.160.14'
12	sInterfaceID	HW_ANY	262



5.1.2. Block description

IEC61850_COM block handles the IEC 61850 protocol. It manages all connected tag blocks. It recognizes cyclic read tags, creates read requests and dispatches the values to the connected blocks. Furthermore, reports are dispatched to the dedicated blocks.

In the following, the most important inputs and outputs will be explained in detail.

RED_CONNECT

Do not change the default value of this input (default value 16#8001). This input is just necessary for H-systems and these systems will not be supported by TIA Portal.

CON_ID

CONN_ID specify the used connection between the PLC or CP and the IEC 61850 device. Every IEC61850_COM or IEC61850_RCOM block must have a unique CON_ID.

MODE

Mode input configures the way the IEC61850_COM block requests the cyclically read values:

- **DEBUG_MODE (0)** means, the IEC_61850_COM block is running in debug mode. It sends a read request for every tag, which should be read cyclically and evaluates its response. In that mode, it can be verified if an IEC 61850 tag address is not set up correctly. (QBAD = TRUE).

•

NOTE

This mode is for debugging only.

NOTE

This mode does not support IEC_CTRL, IEC_REPORT. Although the tag blocks can be connected to the IEC61850_COM block, they will not work properly.

•

- **DATASET (1)** stands for DataSet mode. This means that at the beginning a list (DataSet) is created in the IEC 61850 device. When running in ZYKLCOM, all variables are read with requesting the list name. Not all devices support the services to create a DataSet dynamically over the IEC 61850 protocol but allow configuring a DataSet with an Engineering Tool.
- **LIST (2)** mode reads all the connected tags within one request. Therefore, a list of tags containing all addresses is read cyclically. This mode is very scan time consuming. It should only be used for devices which don't support DataSets.
- Maybe the TIMEOUT variable has to be increased. This is indicated if the IEC61850_COM block reaches the "ZYKLCOM" state but does not keep it stable.

The following configurations are possible:

Table 5-1 DATASET configuration

Input name	DataSet supported		No DataSets
	Dynamic DataSets	Static DataSets	Device does not support DataSets, use LIST mode.
	Device allows dynamic creation of DataSets.	DataSets can only be created with engineering tool.	
MODE	1 = DataSet	1 = DataSet	2 = List
DS_CONF_OP	1 = Supported	0 = Not Supported	0 = Not Supported
DS_NAME	DataSet name of your choice. Will be created in device automatically by the library. e.g., "CTRL/LLNO\$\$DataSet1".	Enter the DataSet name, which was previously configured with the external engineering tool, e.g., „CTRL/LLNO\$\$DataSet1"	-

•

NOTE

It is recommended to set MODE to DATASET, due to the fact it is the fastest way to poll tags cyclically. If your IEC 61850 device does not support datasets, use LIST mode.

NOTE

DEBUG mode should only be used to detect address configuration faults.

NOTE

LIST mode as well as DATASET mode require a RESET for engineering changes (add, delete blocks, address change).

Please see also table DATASET configuration at chapter 3.4.1.

IED_NAME

The IED name has to be set with this input (IED: Intelligent Electronic Device). The IED name is a string and is set up with your engineering tool. (E.g., IED_001).

It can be read out with the IEC 61850 browser tool.

DS_CONF_OP

DS_CONF_OP stands for DataSet Configuration Operations. These can either be supported or not supported. (Dynamic create, delete DataSets). Check your device description to see if these operations are supported.

DS_NAME

The DataSet name is a string to identify your DataSet to cyclically read tags. The DataSet is placed in the IEC 61850 device VMD area (entering only the DataSet name like "DS1" or in the Device Domain/Item area, like "CTRL/LLNO\$\$DS1").

NOTE

The configured tag blocks have to be engineered in the same order as in the DataSet. DataSet entries must not contain objects, but only attributes, except for ALARM tags.

RP_CONF_OP

RP_CONF_OP stands for Reporting Configuration Operation. If the IEC 61850 device supports subscribing and unsubscribing reports, select "SUPPORTED", otherwise select "NOT_SPTED".

If subscribing to reports is selected, the IEC61850_COM block enables the buffered reporting for all tags included in the DataSet specified with RP_DSADDR (whole address) with the trigger "on change". Therefore, a free Report Control block address which contains the RP_DSADDR in entry DataSet has to be specified in RP_CTRL_ADDR.

If RP_CONF_OP is set to "SUPPORTED", each installed report block (IEC_REP) will be initialized during communication startup with the values of the connected IEC 61850-Server device. After initialization, report values will only be changed if a report is triggered through a connected IEC 61850-Server device.

RP_DSADDR

Specify the DataSet containing the tags for reporting. The whole DataSet address has to be configured. E.g., 'CTRL/LLNO\$\$DataSet1'.

NOTE

It is recommended to **not** configure the same DataSet you have configured on input DS_NAME for cyclically reading the tag blocks (e.g., IEC_RD_BO, IEC_RD_IN...). For reporting functionality, you have to create a separate DataSet within your IEC 61850 server device with the related IEC 61850 device engineering tool.

Further enter the Report Control block address (e.g., "CTRL/LLNO\$\$BR\$\$brcbA01") at the input RP_CTRL_ADDR.

RP_CTRL_ADDR

Specify a free Report Control block address to enable reporting. Normally, the corresponding Report Control blocks reside in the same tree part as the Report DataSet. E.g., 'CTRL/LLNO\$\$BR\$\$brcbA01'.

IP_ADDR

Enter the IEC 61850 server IP address (protection relay) when using T- communication.

TSAP_RID

Enter the IEC 61850 server TSAP number (default no: 0001 when using T- communication).

Interface_ID

Hardware identifier of the local interface. Find this ID in the properties of your communication module.

SAMPLE_T

Use SAMPLE_T to write the block OB cycle time in ms/100. This input should never be 0.0. It is used for TIMEOUT calculation.

EN_MSG

Enables WinCC messaging if set to "1".

STRT_COM

OFF deactivates the communication via IEC 61850. ON starts the communication again.

RESET

Reset deactivates the IEC 61850 communication, sets all internal variables to INIT value and re-organizes the internal block list. It has to be executed, if for example a tag block is removed or added, or an address has changed.

CONNECT

CONNECT output has to be connected with the tag block RO_CONNECT input.

QERR

QERR is set to "1" if any errors occur.

QERR_NO

QERR_NO delivers detailed error information. (Not yet supported.)

QRACKF

QRACKF is set to "1" if a RACK failure occurs.

QPARF

QPARF is set to "1" if the plausibility check of connection parameters LADDR and CONN_ID failed.

QCOMF

QCOMF is set to "1" if the IEC 61850 communication is not in state "ZYKLCOM". Communication error.

QB_CR_DS

If QB_CR_DS is set to "1", the DataSet with the name specified in DS_NAME could not be created.

QB_EN_RP

If QB_EN_RP is set to "1", writing the Report configuration failed. Reporting could not be enabled.

NOTE

If QB_EN_RP is set to "1", all connected IEC_REPORT blocks will not work properly.

QSTATE

IEC 61850_COM is mainly organized with a state machine. QSTATE shows the current state. If communication to the IEC 61850 device is established correctly, the function block is running in state "ZYKLCOM" after initialization.

The following table lists the possible states:

Table 5-2 IEC61850_COM QSTATE output

QSTATE	PROG_STATE	Description
0	'INIT'	Initialization state to get parameter
1	'CONREQ'	Send initiate request
2	'CONRES'	Receive initiate response
3	'CONERROR'	Connection error (please see table on next page)
4	'CDSEXREQ'	Send Request to check existence of DataSet
5	'CDSEXRES'	Receive Response for DataSet existence request
6	'CDSCRREQ'	Create DataSet request
7	'CDSCRRES'	Create Dataset response
8	'CSDLREQ'	Delete DataSet request
9	'CSDLRES'	Delete DataSet response
10	'CINFOREQ'	Write ReportCtrlblock request
11	'CINFORES'	Write ReportCtrlblock response
12	'ZYKLCOM'	Cyclic communication state.
13	'CONCLREQ'	Conclude Request
14	'CONCLRES'	Conclude Response
15	'FINREQ'	Release Request
16	'FINRES'	Release Response
17	'ABORT'	Abort received

5.1.3. Called blocks

Table 5-3 IEC61850_COM called blocks

Declaration/Block number	Declaration/Block name
FB501	MMS_EncLen
FB502	MMS_DecLen
FB503	MMS_Enc_TLV
FB504	MMS_Dec_TLV
FB505	OSI_Dec_Prot
FB507	MMS_Declnt
FB511	MMS_BuildReq
FB512	MMS_ParseResp
FB513	IEC61850CLPgrLgc
FB514	IEC61850_Link
FB529	MMS_ParseDSAddr
FB531	MMS_Strin_IP_Konv
FB700	Program_Alarm
FB1030	TSEND_C
FB1031	TRCV_C
N/A	BST_BOOL_SHORT

Declaration/Block number	Declaration/Block name
N/A	BST_INT_SHORT
N/A	BST_REAL_SHORT
N/A	BST_CTRL_SHORT
N/A	IEC61850_Params
N/A	BST_BSTR_SHORT
N/A	BST_SHORT
N/A	LINK_INFO
N/A	LEN
N/A	BLKMOV
N/A	LGC_GADR
N/A	RD_SINFO
N/A	TCON
N/A	TDISCON

5.1.4. I/O description

Depending on the used library, not all inputs and outputs in the list will still be required. These inputs and outputs wouldn't exist on the block.

NOTE

These inputs and outputs are marked in the list with the extension (opt.).

Table 5-4 IEC61850 I/O description

I/O	I/O Name	Type	I/O Default	Comment
I	RED_CONNECT	WORD	16#8001	Connection to IEC61850_RED block
I	CON_ID	INT	0	connection ID
I	TIMEOUT	DINT	1000	Timeout in milliseconds
I	MODE	INT	0	Debug mode (0) / DATASET (1) / LIST (2)
I	IED_NAME	STRING	XXX	IEC 61850 IED name
I	DS_CONF_OP	INT	0	DataSet Configuration Operation SUPPORTED / NOT SUPPORTED
I	DS_NAME	STRING	XXX	IEC 61850 Dataset name
I	RP_CON_OP(opt.)	INT	0	Report Configuration Operation SUPPORTED / NOT SUPPORTED
I	RP_DSADDR(opt.)	STRING	XXX	DataSet address used for Reporting
I	RP_CTRL_ADDR(opt.)	STRING	XXX	DataSet Control block address
I	MSG_EVID(opt.)	DWORD	2	Alarm_8P message ID // deprecated
I	SAMPLE_T	REAL		Sample time input (system input)
I	EN_MSG(opt.)	BOOL	0	1 = enable alarm
I	MSG_EVID(opt.)	BOOL	0	System input for Alarming
I	IP_ADDR(opt.)	STRING	0	IEC 61850 device IP address
I	TSAP_RID(opt.)	WORD	0	TSAP remote ID for T-communication
I	Interface_ID	HW_ANY	64	ID of Interface
I	CPU_ID(opt.)	BYTE	0	Select correct CPU/CP for T-communication
I	TCON_ID(opt.)	WORD	0	Use unique number for T-communication instance
IO	STRT_COM	BOOL	0	1 = start communication

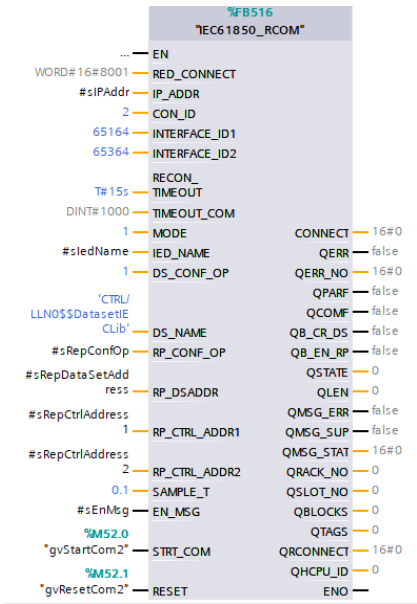
I/O	I/O Name	Type	I/O Default	Comment
IO	RESET	BOOL	0	1 = reset function block
O	CONNECT	BOOL	1	Connection to input R_CONNECT of READ_IEC blocks
O	QERR	BOOL	0	General error
O	QERR_NO	WORD	0	Error number – not yet supported
O	QRACKF	BOOL	0	1 = Rack failure (power fault, CPU error, ...)
O	QPARF	BOOL	0	1 = Parameter assignment error module
O	QCOMF	BOOL	0	1 = Communication error, Communication not running in ZYKLCOM.
O	QB_CR_DS	BOOL	1	Create DataSet failed
O	QB_EN_RP(opt.)	BOOL	1	Subscribing Report failed
O	QSTATE	INT	0	Internal state
O	QSNDACT	BOOL	0	1 = data sent
O	QSNDDIAG	WORD	0	Detailed send status
O	QRCVDAT	BOOL	0	1 = data received
O	QRCVDIAG	WORD	0	Detailed receive status
O	QLEN	INT	0	Length of received data
O	QMSG_ERR(opt.)	BOOL	0	1 = Message Error
O	QMSG_SUP(opt.)	BOOL	0	1 = Message suppression active
O	MSG_STAT(opt.)	WORD	0	ALARM_8P: STATUS Output
O	MSG_ACK(opt.)	WORD	0	ALARM_8P: ACKNOWLEDGE Output
O	QRACK_NO	INT	0	CP Rack-Number
O	Q SLOT_NO	INT	0	CP Slot-Number
O	QBLOCKS	INT	0	Number of connected blocks
O	QTAGS	INT	0	Number of connected tags
O	QRCONNECT	WORD	16#00	Red. Communication output

5.2. IEC61850_RCOM

5.2.1. Block screenshot

Figure 5-2 IEC61850_RCOM block S7-1500

7	Static		
8	sIedName	String[24]	'IED_0002'
9	sEnMsg	Bool	true
10	sEnTCON	Bool	true
11	sIPAddr	String[15]	'140.80.160.14'
12	sInterfaceID	HW_ANY	262



5.2.2. Block description

IEC61850_RCOM block handles the IEC 61850 protocol. It manages all connected tag blocks and redundancy. It recognizes cyclic read tags, creates read requests and dispatches the values to the connected blocks. Furthermore, reports are dispatched to the dedicated blocks.

In the following, the most important inputs and outputs will be explained in detail.

RECON_TIMEOUT

The input RECON_TIMEOUT defines the time period after a connection reset will be performed in case of a connection interrupt. The shorter the duration is parameterized, the shorter the reconnection time will be after a redundancy switch over. If the parameterized duration is too short, no connection to the IED can be established.

RP_CTRL_ADDR1/2

RP_CTRL_ADDR1/2 specify a free Report Control block address to enable reporting. Normally, the corresponding Report Control blocks reside in the same tree part as the Report DataSet. E.g., 'CTRL/LLNO\$\$BR\$\$brcbA01'.

NOTE

In case of DATASET mode and reporting mechanism, an additional report control block is required. Set an IEC 61850 path to report control block at input RP_CTRL_ADDR2 which is different to RP_CTRL_ADDR1.

Interface_ID1/2

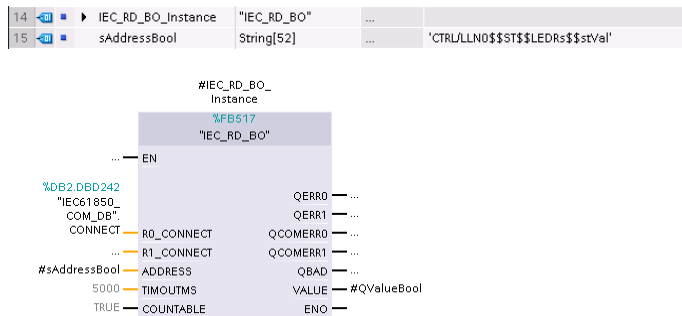
Hardware identifier of the local interface. Find those ID for both redundant PLCs in the properties of your communication module.

Input/Output description is equal to IEC61850_COM Block description. Please see chapter 5.1.

5.3. IEC_RD_BO (BOOL tag)

5.3.1. Block screenshot

Figure 5-3 IEC_RD_BO



5.3.2. Block description

IEC_RD_BO reads one Boolean tag cyclically.

R0_CONNECT

R0_CONNECT input has to be connected to the IEC61850_COM block CONNECT output.

ADDRESS

The address for the tag has to be configured with the input ADDRESS. The address has to be entered as an IEC 61850 address string including the IEC 61850 objects attribute name.

NOTE

The additional character "\$" has to be replaced with "\$\$" in STEP 7

NOTE

The IEC 61850 address strings can also be generated and copied via the IEC Browser.

The function block outputs can be connected with other SIMATIC S7 blocks.

Example for an address: "CTRL/LLN0\$\$ST\$\$LEDrs\$\$stVal".

TIMOUTMS

TIMOUTMS specifies the time in ms after the quality output QBAD changes to true, if there is no actualization of the value.

QERRO

QERRO is set, if the connection to the IEC61850_COM block cannot be found or is faulty.

QCOMERRO

QCOMERRO is set if the IEC 61850 communication handled by the IEC61850_COM is not working properly. IEC61850_COM: QSTATE != ZYKLCOM.

QBAD

QBAD is set if the value is not valid anymore.

Possible reasons are:

- No connection to any IEC61850_COM block
- No IEC 61850 communication running
- Address string not set properly; value was not updated for time TIMOUTMS

VALUE

VALUE represents the actual BOOL value addressed through IEC61850_COM IED_NAME and IEC_RD_BO ADDRESS string.

5.3.3. Called blocks

Table 5-5 IEC_RD_BO called blocks

Declaration/Block name	
BST_BOOL	
LINK_INFO	
LEN	
LEN	
TIME_TCK	

5.3.4. I/O description

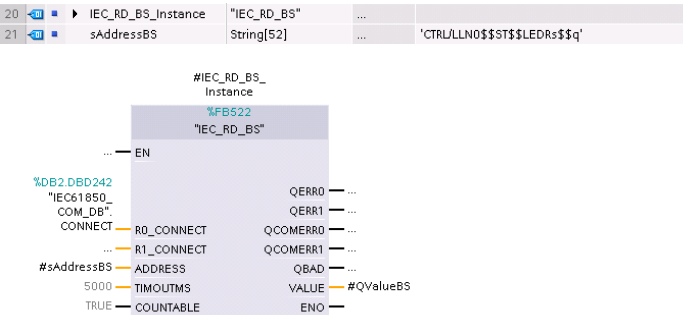
Table 5-6 IEC_RD_BO I/O description

I/O	I/O Name	Type	I/O Default	Comment
I	RO_CONNECT	ANY(Struct)		Connection to IEC61850_COM block DB
I	ADDRESS	STRING[52]		IEC 61850 tag address
I	TIMEOUTS	INT	5000	Timeout in ms
I	COUNTABLE	BOOL	1	Internal value
O	QERRO	BOOL	1	1 = Connection to IEC61850_COM block 0 faulty
O	QCOMERRO	BOOL	1	Communication IEC61850_COM block 0 not ok
O	QBAD	BOOL	1	0 = Data valid
O	VALUE	BOOL	0	Addressed value as BOOL

5.4. IEC_RD_BS (BITSTRING tag)

5.4.1. Block screenshot

Figure 5-4 IEC_RD_BS



5.4.2. Block description

IEC_RD_BS reads one Bitstring tag cyclically. The Bitstring length is limited to 4 Bytes. Bitstrings with a length bigger than 4 bytes cannot be parsed.

Input/Output description is equal to IEC_RD_BO Block description. Please see chapter 5.3.2.

5.4.3. Called blocks

Table 5-7 IEC_RD_BS called blocks

Declaration/Block name
BST_BSTR
LINK_INFO
EQ_STRING
LEN
BLKMOV
TIME_TCK

5.4.4. I/O description

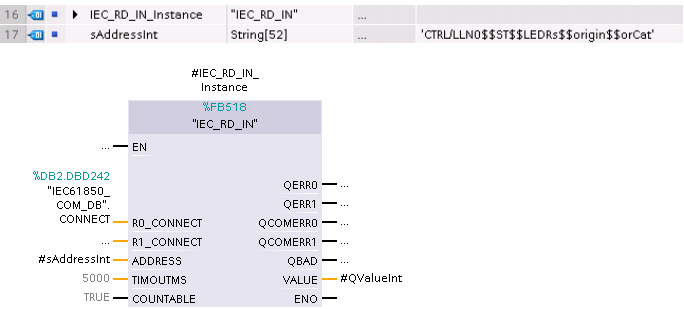
Table 5-8 IEC_RD_BS I/O description

I/O	I/O Name	Type	I/O Default	Comment
I	RO_CONNECT	ANY(Struct)		Connection to IEC61850_COM block DB
I	ADDRESS	STRING[52]		IEC 61850 tag address
I	TIMEOUTS	INT	5000	Timeout in ms
I	COUNTABLE	BOOL	1	Internal value
O	QERRO	BOOL	1	1 = Connection to IEC61850_COM block 0 faulty
O	QCOMERRO	BOOL	1	Communication IEC61850_COM block 0 not ok
O	QBAD	BOOL	1	0 = Data valid
O	VALUE	DWORD	0	Addressed value as BITSTRING

5.5. IEC_RD_IN (INTEGER tag)

5.5.1. Block screenshot

Figure 5-5 IEC_RD_IN



5.5.2. Block description

IEC_RD_IN reads one DINT tag cyclically. The Integer size is 32 bit (4byte).

Input/Output description is equal to the IEC_RD_BO Block description. Please see chapter 5.3.2.

5.5.3. Called blocks

Table 5-9 IEC_RD_IN called blocks

Declaration/Block name
BST_INT
LINK_INFO

Declaration/Block name
EQ_STRING
LEN
BLKMOV
TIME_TCK

5.5.4. I/O description

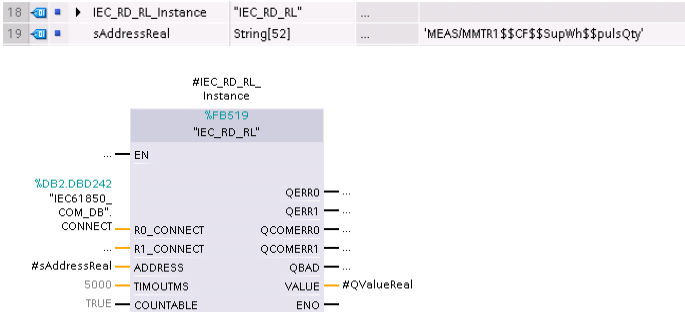
Table 5-10 IEC_RD_IN I/O description

I/O	I/O Name	Type	I/O Default	Comment
I	RO_CONNECT	ANY(Struct)		Connection to IEC61850_COM block DB
I	ADDRESS	STRING[52]		IEC 61850 tag address
I	TIMEOUTS	INT	5000	Timeout in ms
I	COUNTABLE	BOOL	1	Internal value
O	QERRO	BOOL	1	1 = Connection to IEC61850_COM block 0 faulty
O	QCOMERRO	BOOL	1	Communication IEC61850_COM block 0 not ok
O	QBAD	BOOL	1	0 = Data valid
O	VALUE	DINT	0	Addressed value as Integer (DINT)

5.6. IEC_RD_RL (REAL tag)

5.6.1. Block screenshot

Figure 5-6 IEC_RD_RL



5.6.2. Block description

IEC_RD_RL reads one REAL tag cyclically. The Floating point value size is 32 bit (4byte).

Input/Output description is equal to IEC_RD_BO Block description. Please see chapter 5.3.2.

5.6.3. Called blocks

Table 5-11 IEC_RD_RL called blocks

Declaration/Block name
BST_REAL
LINK_INFO
EQ_STRING
LEN
BLKMOV

Declaration/Block name
TIME_TCK

5.6.4. I/O description

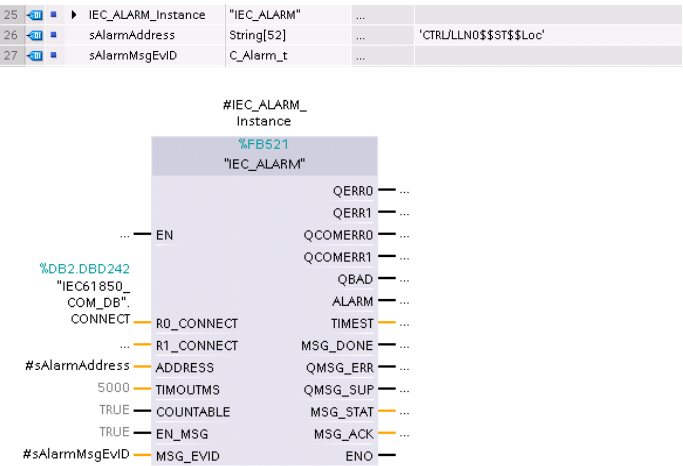
Table 5-12 IEC_RD_RL I/O description

I/O	I/O Name	Type	I/O Default	Comment
I	RO_CONNECT	ANY(Struct)		Connection to IEC61850_COM block DB
I	ADDRESS	STRING[52]		IEC 61850 tag address
I	TIMEOUTS	INT	5000	Timeout in ms
I	COUNTABLE	BOOL	1	Internal value
O	QERRO	BOOL	1	1 = Connection to IEC61850_COM block 0 faulty
O	QCOMERRO	BOOL	1	Communication IEC61850_COM block 0 not ok
O	QBAD	BOOL	1	0 = Data valid
O	VALUE	REAL	0	Addressed value as Floating point value (REAL)

5.7. IEC_ALARM (BOOL tag with timestamp)

5.7.1. Block screenshot

Figure 5-7 IEC_ALARM



5.7.2. Block description

IEC_ALARM reads one BOOL value cyclically with its timestamp, which is updated if the variable changes.

NOTE

The IEC 61850 address strings can also be generated and copied via the IEC Browser. Example for an address: "CTRL/LLN0\$\$ST\$\$LEDs"

NOTE

The address has to be set without the IEC 61850 object's attribute name, only the IEC 61850 object name has to be set.

EN_MSG

EN_MSG enables Alarming in WinCC. Special messages can be specified.

Further Input/Output description is equal to the IEC_RD_BO Block description. Please see chapter 5.3.2.

5.7.3. Called blocks

Table 5-13 IEC_ALARM called blocks

Declaration/Block name
UTCcalc
ALARM_8P
BST_BOOL
LINK_INFO
EQ_STRING
LEN
BLKMOV
TIME_TCK

5.7.4. I/O description

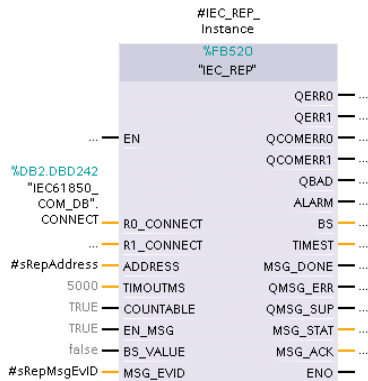
Table 5-14 IEC_ALARM I/O description

I/O	I/O Name	Type	I/O Default	Comment
I	RO_CONNECT	ANY(Struct)		Connection to IEC61850_COM block DB
I	ADDRESS	BOOL	1	IEC 61850 tag address
I	TIMEOUTS	INT	5000	Timeout in milliseconds
I	COUNTABLE	BOOL	1	Internal value
I	EN_MSG			1 = enable alarming in WinCC
O	QERRO	BOOL	1	1 = Connection to IEC61850_COM block 0 faulty
O	QCOMERRO	BOOL	1	Communication IEC61850_COM block 0 not ok
O	QBAD	BOOL	1	0 = Data valid
O	ALARM	REAL	0	Alarm value (BOOL)
O	TIMEST	DT		Time Stamp of Value
O	MSG_DONE	BOOL	0	1 = ALARM_8 DONE
O	QMSG_ERR	BOOL	0	1 = ALARM_8 Error
O	QMSG_SUP	BOOL	0	1 = Message suppression active
O	MSG_STAT	WORD	0	ALARM_8P: STATUS Output
O	MSG_ACK	WORD	0	ALARM_8P: STATUS Output

5.8. IEC_REP (Report – BOOL tag with timestamp)**5.8.1. Block screenshot**

Figure 5-8 IEC_REP

22	IEC_REP_Instance	"IEC_REP"	...	
23	sRepAddress	String[52]	...	'CTRL/LLN0\$\$\$ST\$\$Loc'
24	sRepMsgEvid	C_Alarm_t	...	



5.8.2. Block description

IEC_Rep reads one BOOL value asynchronous via IEC 61850 Report mechanism with its timestamp.

To use this Report block, the IEC 61850 device has to support the IEC 61850 service Report. Furthermore, a DataSet containing the tag (addressed with IEC_Rep) has to exist in the device. The DataSet name and its address for Reporting have to be set in IEC61850_COM: RP_DSADDR. Furthermore, enabling Reporting has to be executed successfully. IEC61850_COM: QB_EN_RP has to be "0" after initialization.

ADDRESS

The address has to be set without the IEC 61850 object's attribute name, only the IEC 61850 object name has to be set.

NOTE

The IEC 61850 address strings can also be generated and copied via the IEC Browser. Example for an address: "CTRL/LLNO\$\$ST\$\$LEDRs"

EN_Msg

EN_Msg enables Alarming in WinCC. Special messages can be specified.

Further Input/Output description is equal to IEC_RD_BO Block description. Please see chapter 5.3.2.

NOTE

When adding two IEC_Rep tag blocks with the same address, only the first IEC_Rep tag will be updated. The library allows no duplicate IEC_Rep blocks.

NOTE

It is not possible to use the block for complex structures (structures which contain more than one bool value)

5.8.3. Alarming

There two different messages being triggered, depending on the value data type (BOOL or BITSTRING). For both messages, the timestamps of the IEC 61850 report event (e.g., switch change time in IEC 61850 server device) are delivered to WinCC.

Table 5-15

Message no.	Block parameters	Default message text	Message class	Message no.
1	ALARM	Report Alarm Message 1	ALARM – high	1

5.8.4. Called blocks

Table 5-16 IEC_REP called blocks

Declaration/Block name
BST_REP
LINK_INFO
EQ_STRING
LEN
BLKMOV
TIME_TCK

5.8.5. I/O description

Table 5-17 IEC_REP I/O description

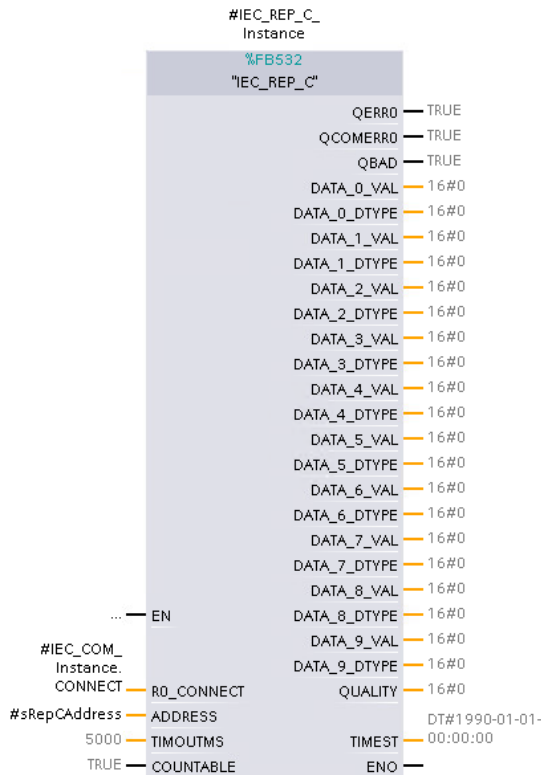
I/O	I/O Name	Type	I/O Default	Comment
I	R0_CONNECT	ANY(Struct)		Connection to IEC61850_COM block DB
I	ADDRESS	BOOL	1	IEC 61850 tag address
I	TIMEOUTS	INT	5000	Timeout in milliseconds
I	COUNTABLE	BOOL	1	Internal value
I	EN_MSG	BOOL	1	1 = enable alarming in WinCC
I	BS_VALUE	BOOL	0	1 = value with BITRSTING type
O	QERRO	BOOL	1	1 = Connection to IEC61850_COM block 0 faulty
O	QCOMERRO	BOOL	1	Communication IEC61850_COM block 0 not ok
O	QBAD	BOOL	1	0 = Data valid
O	ALARM	REAL	0	Alarm value (BOOL)
O	TIMEST	DT		Time Stamp of Value
O	MSG_DONE	BOOL	0	1 = ALARM_8 DONE
O	QMSG_ERR	BOOL	0	1 = ALARM_8 Error
O	QMSG_SUP	BOOL	0	1 = Message suppression active
O	MSG_STAT	WORD	0	ALARM_8P: STATUS Output
O	MSG_ACK	WORD	0	ALARM_8P: STATUS Output

5.9. IEC_REP_C (Report – complex structure with timestamp)

5.9.1. Block screenshot

41	IEC_REP_C_Instance	"IEC_REP_C"	
42	sRepCAddress	String[52]	'PROT/PTOC12\$\$ST\$\$Str'

Figure 5-9 IEC_REP_C



5.9.2. Block description

IEC_REP_C reads a complex structure with up to 10 values asynchronous via IEC 61850 Report mechanism with its timestamp.

To use this Report block, the IEC 61850 device has to support the IEC 61850 service Report. Furthermore, a DataSet containing the structure (addressed with IEC_REP_C) has to exist in the device. The DataSet name and its address for Reporting have to be set in IEC61850_COM: RP_DSADDR. Furthermore, enabling Reporting has to be executed successfully. IEC61850_COM: QB_EN_RP has to be "0" after initialization.

Compared to the IEC_REP block, the IEC_REP_C does not support alarming. In addition, the transferred data points of the structure are output as raw values (at output DATA_X_VAL) and may still need to be converted. For every data point the datatype is displayed at the output DATA_X_DTYPE. The following table shows which value corresponds to which data type.

DATA_X_DTYPE Value	Datatype
16#83	Bool
16#84	BitString
16#85	Integer
16#86	UInteger
16#87	Real
16#91	UTC_Time

Table 5-18 IEC_REP_C data type

ADDRESS

The address has to be set without the IEC 61850 object’s attribute name, only the IEC 61850 object name has to be set.

NOTE

The IEC 61850 address strings can also be generated and copied via the IEC Browser.
Example for an address: “PROT/PTOC1\$\$ST\$\$Str”

Further Input/Output description is equal to IEC_RD_BO Block description. Please see chapter 5.3.2.

NOTE

When adding two IEC_REP_C tag blocks with the same address, only the first IEC_REP_C tag will be updated. The library allows no duplicate IEC_REP_C blocks.

5.9.3. Called blocks

Declaration/Block name
BST_REP
LINK_INFO
EQ_STRING
LEN
BLKMOV
TIME_TCK

Table 5-19 IEC_REP_C called blocks

5.9.4. I/O description

I/O	I/O Name	Type	I/O Default	Comment
I	RO_CONNECT	ANY(Struct)		Connection to IEC61850_COM block DB
I	ADDRESS	BOOL	1	IEC 61850 tag address
I	TIMEOUTS	INT	5000	Timeout in milliseconds
I	COUNTABLE	BOOL	1	Internal value
O	QERR0	BOOL	1	1 = Connection to IEC61850_COM block 0 faulty
O	QCOMERR0	BOOL	1	Communication IEC61850_COM block 0 not ok
O	QBAD	BOOL	1	0 = Data valid
O	DATA_X_VAL	DWord	0	Data value (raw value)
O	DATA_X_DTYPE	BYTE	0	Data type of value
O	TIMEST	DT		Time Stamp of Value

Table 5-20 IEC_REP_C I/O description

5.10. C_CTRL

5.10.1. Block screenshot

Figure 5-10 IEC_CTRL



5.10.2. **Block description**

IEC_CTRL controls an IEC 61850 device switch using the **IEC 61850 Control Service**. If CTRL is set to "1", a Control command is executed with the Control value CTRLVAL.

ADDRESS

The address has to be set without the IEC 61850 object's attribute name, only the IEC 61850 object name has to be set. Addresses for Control objects contain the function constraint "CO".

NOTE

The IEC 61850 address strings can also be generated and copied via the IEC Browser.
Example for an address: "CTRL/LLN0\$\$CO\$\$LEDRs"

ORIGIN

Specify the command source (Actor). This string cannot be empty. Default value is "SIEMENS".

WITHILOC

If WITHILOC is set before executing control command, an Interlock check is done in the device.

WITHSYNC

If WITHSYNC is set before executing control command, a Synchro check is done in the device.

TEST

If TEST is set, the switch command is only for testing, real switching is not executed.

CTRLVAL

Specifies the Command value: "0" stands for close, "1" represents open.

SBO

Specifies if "Select before Operate" should be used. If "Select before Operate" is selected, a "Select" telegram is sent to the IEC 61850 device first and after acknowledging, a second "Operate" telegram is sent to control the switch. If SBO is disabled, only an "Operate" telegram is sent to the device.

QBAD

If the DONE bit is not raised up for time TIMOUT, QBAD gets "1", else QBAD equals "0".

DONE

If the switch execution was successful, the DONE bit is raised up.

Further Input/Output description is equal to IEC_RD_BO Block description. Please see chapter 5.3.2.

5.10.3. **Called blocks**

Table 5-21 IEC_CTRL called blocks

Declaration/Block name
BST_CTRL
LINK_INFO
EQ_STRING
LEN
BLKMOV
TIME_TCK

5.10.4. I/O description

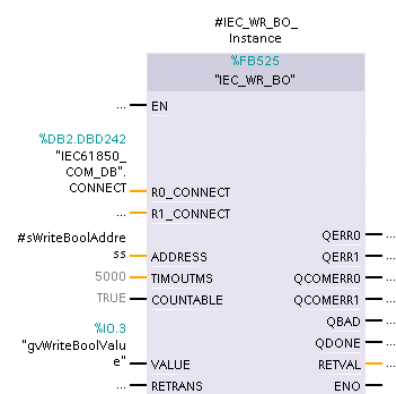
Table 5-22 IEC_CTRL I/O description

I/O	I/O Name	Type	I/O Default	Comment
I	RO_CONNECT	ANY(Struct)		Connection to IEC61850_COM block DB
I	ADDRESS	BOOL	1	IEC 61850 tag address
I	TIMEOUTS	INT	5000	Timeout in milliseconds
I	COUNTABLE	BOOL	1	Internal value
I	ORIGIN	STRING[52]		Actor name, control source
I	WITHLOCK			1 = enables Interlock check
I	WITHSYNCHR			1 = enables Synchro check
I	TEST			Only test switch control command
I	CTRLVAL			Switch command 1: open 0: close
I	SBO			1: Select before operate 0: only operate
IO	CTRL	BOOL		Control command trigger, is reset automatically
O	QERR0	BOOL	1	1 = Connection to IEC61850_COM block 0 faulty
O	QCOMERR0	BOOL	1	Communication IEC61850_COM block 0 not ok
O	QBAD	BOOL	1	0: QDONE raised up after CTRL = 1 1: QDONE did not raise up after CTRL = 1
O	QDONE	REAL	0	Switch operation executed successfully

5.11. IEC_WR_BO

5.11.1. Block screenshot

Figure 5-11 IEC_WR_BO



5.11.2. Block description

With the IEC_WR_BO block a Boolean variable can be written to an IEC 61850 Server Device. This block can be used for example to write substitution values.

R0_CONNECT

R0_CONNECT input has to be connected to the IEC61850_COM block CONNECT output.

ADDRESS

The address for the tag has to be configured with the input ADDRESS. The address has to be entered as an IEC 61850 address string including the IEC 61850 object's attribute name.

NOTE

The additional character "\$" has to be replaced with "\$\$" in STEP 7

NOTE

The IEC 61850 address strings can also be generated and copied via the IEC Browser.
Example for an address: "CTRL/LLN0\$\$ST\$\$LEDRs\$\$stVal".

TIMOUTMS

TIMOUTMS specifies the time in ms after the output QBAD changes to true, if the IEC 61850 device does not respond with a positive acknowledge in between.

VALUE

Represents the Boolean value which should be written to the IEC 61850 device. Write is performed on value change.

RETRANS

If RETRANS is set to "TRUE", the current value will be retransmitted. Input resets automatically to "FALSE".

QERRO

QERRO is set if the connection to the IEC61850_COM block cannot be found or is faulty.

QCOMERRO

QCOMERRO is set if the IEC 61850 communication the IEC61850_COM handles is not working properly.
IEC61850_COM: QSTATE != ZYKLCOM.

QBAD

QBAD is set if the value is not valid anymore.

Possible reasons are:

- No connection to any IEC61850_COM block
- No IEC 61850 communication running
- Address string not set properly, value was not written successfully in between specified timeout
- Access rights: no permission for writing value / value cannot be overwritten

DONE

Done will be set to TRUE, after transmission of value

RETV

RETV shows the IEC 61850 Write return value.

5.11.3. Called blocks

Table 5-23 IEC_WR_BO called blocks

Declaration/Block name
BST_BOOL
LINK_INFO
EQ_STRING
LEN
BLKMOV
TIME_TCK

5.11.4. I/O description

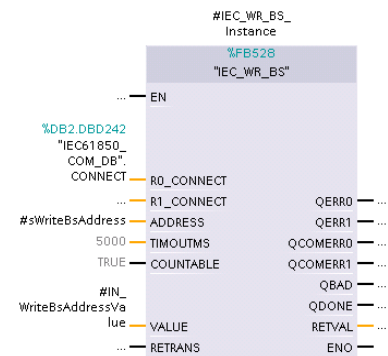
Table 5-24 IEC_WR_BO I/O description

I/O	I/O Name	Type	I/O Default	Comment
I	RO_CONNECT	ANY(Struct)		Connection to IEC61850_COM block DB
I	ADDRESS	STRING[52]		IEC 61850 tag address
I	TIMEOUTS	INT	5000	Timeout in ms
I	COUNTABLE	BOOL	1	Internal value
I	VALUE	BOOL	1	Tag value to be written in device
I	RETRANS	BOOL	1	Retransmission of value
O	QERRO	BOOL	1	1 = Connection to IEC61850_COM block 0 faulty
O	QCOMERRO	BOOL	1	Communication IEC61850_COM block 0 not ok
O	QBAD	BOOL	1	0 = Data valid
O	QDONE	BOOL	0	Addressed value as BITSTRING
O	RETVAl	BYTE	0	Return code

5.12. IEC_WR_BS

5.12.1. Block screenshot

Figure 5-12 IEC_WR_BS



5.12.2. Block description

With the IEC_WR_BS block a Bitstring variable can be written in an IEC 61850 Server Device. This block can be used for example to write substitution values.

RO_CONNECT

RO_CONNECT input has to be connected to the IEC61850_COM block CONNECT output.

ADDRESS

The address for the tag has to be configured with the input ADDRESS. The address has to be entered as an IEC 61850 address string including the IEC 61850 object's attribute name.

NOTE

The additional character "\$" has to be replaced with "\$\$" in STEP 7

NOTE

The IEC 61850 address strings can also be generated and copied via the IEC Browser.
Example for an address: "CTRL/LLN0\$\$ST\$\$LEDs\$\$stVal".

TIMOUTMS

TIMOUTMS specifies the time in ms after the output QBAD changes to true, if the IEC 61850 device does not respond with a positive acknowledge in between.

MODE

MODE specifies whether a Bitstring value (e.g. \$subVal) or quality (e.g. \$subQ) is written. If MODE is set to "FALSE" (standard value) a value can be written. For a quality code MODE must be set to "TRUE".

VALUE

Represents the Boolean value which should be written to the IEC 61850 device. Write is performed on value change.

RETRANS

If RETRANS is set to "TRUE", the current value will be retransmitted. Input resets automatically to "FALSE".

QERRO

QERRO is set if the connection to the IEC61850_COM block cannot be found or is faulty.

QCOMERRO

QCOMERRO is set if the IEC 61850 communication the IEC61850_COM handles is not working properly.
IEC61850_COM: QSTATE != ZYKLCOM.

QBAD

QBAD is set if the value is not valid anymore.

Possible reasons are:

- No connection to any IEC61850_COM block
- No IEC 61850 communication running
- Address string not set properly, value was not written successfully in between specified timeout
- Access rights: no permission for writing value / value cannot be overwritten

DONE

Done will be set to TRUE, after transmission of value

RETVAL

RETVAL shows the IEC 61850 Write return value.

5.12.3. Called blocks

Table 5-25 IEC_WR_BS called blocks

Declaration/Block name
BST_BITSTRING
LINK_INFO
EQ_STRING
LEN
BLKMOV
TIME_TCK

5.12.4. I/O description

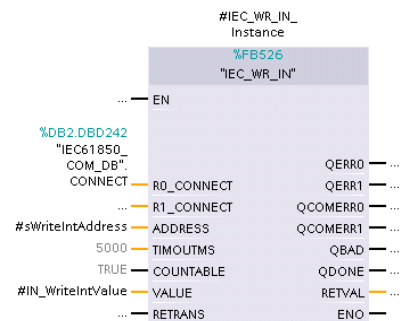
Table 5-26 IEC_WR_BS I/O description

I/O	I/O Name	Type	I/O Default	Comment
I	RO_CONNECT	ANY(Struct)		Connection to IEC61850_COM block DB
I	ADDRESS	STRING[52]		IEC 61850 tag address
I	TIMEOUTS	INT	5000	Timeout in ms
I	COUNTABLE	BOOL	1	Internal value
I	MODE	BOOL	0	0 = value 1= quality to be written
I	VALUE	DWORD	1	Tag value to be written in device
I	RETRANS	BOOL	1	Retransmission of value
O	QERRO	BOOL	1	1 = Connection to IEC61850_COM block 0 faulty
O	QCOMERRO	BOOL	1	Communication IEC61850_COM block 0 not ok
O	QBAD	BOOL	1	0 = Data valid
O	QDONE	BOOL	0	Addressed value as BITSTRING
O	RETVAL	BYTE	0	Return code

5.13. IEC_WR_IN

5.13.1. Block screenshot

Figure 5-13 IEC_WR_IN



5.13.2. Block description

With the IEC_WR_IN block, an Integer variable can be written in an IEC 61850 Server Device. This block can be used for example to write substitution values.

Input/Output description is equal to IEC_WR_BO Block description. Please see chapter 5.11.2.

5.13.3. Called blocks

Table 5-27 IEC_WR_IN called blocks

Declaration/Block name
BST_INT
LINK_INFO
EQ_STRING
LEN
BLKMOV
TIME_TCK

5.13.4. I/O description

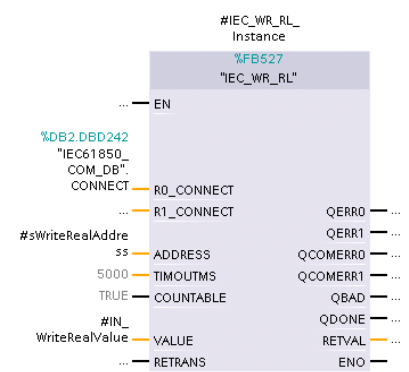
Table 5-28 IEC_WR_IN I/O description

I/O	I/O Name	Type	I/O Default	Comment
I	RO_CONNECT	ANY(Struct)		Connection to IEC61850_COM block DB
I	ADDRESS	STRING[52]		IEC 61850 tag address
I	TIMEOUTS	INT	5000	Timeout in ms
I	COUNTABLE	BOOL	1	Internal value
I	VALUE	DINT	1	Tag value to be written in device
I	RETRANS	BOOL	1	Retransmission of value
O	QERRO	BOOL	1	1 = Connection to IEC61850_COM block 0 faulty
O	QCOMERRO	BOOL	1	Communication IEC61850_COM block 0 not ok
O	QBAD	BOOL	1	0 = Data valid
O	QDONE	BOOL	0	Addressed value as BITSTRING
O	RETV	BYTE	0	Return code

5.14. IEC_WR_RL

5.14.1. Block screenshot

Figure 5-14 IEC_WR_RL



5.14.2. Block description

With the IEC_WR_RL block, a Real variable can be written in an IEC 61850 Server Device. This block can be used for example to write substitution values.

Input/Output description is equal to IEC_WR_BO Block description. Please see chapter 5.11.2.

5.14.3. Called blocks

Table 5-29 IEC_WR_RL called blocks

Declaration/Block name
BST_INT
LINK_INFO
EQ_STRING
LEN
BLKMOV
TIME_TCK

5.14.4. I/O description

Table 5-30 IEC_WR_RL I/O description

I/O	I/O Name	Type	I/O Default	Comment
I	RO_CONNECT	ANY(Struct)		Connection to IEC61850_COM block DB
I	ADDRESS	STRING[52]		IEC 61850 tag address
I	TIMEOUTS	INT	5000	Timeout in ms
I	COUNTABLE	BOOL	1	Internal value
I	VALUE	REAL	1	Tag value to be written in device
I	RETRANS	BOOL	1	Retransmission of value
O	QERRO	BOOL	1	1 = Connection to IEC61850_COM block 0 faulty
O	QCOMERRO	BOOL	1	Communication IEC61850_COM block 0 not ok
O	QBAD	BOOL	1	0 = Data valid
O	QDONE	BOOL	0	Addressed value as BITSTRING
O	RETVAL	BYTE	0	Return code

6. Troubleshooting

Table 6-1 Troubleshooting IEC 61850 communication

No	Problem	Possible Reason	Solution
1.	IEC61850_COM does not get into QSTATE "ZYKLCOM"	MODE: DATASET, some of the tag addresses are not valid	Change mode into LIST, to see which tags keep QBAD = "TRUE". Check tag addresses with the IEC 61850 browser tool.
		MODE: DATASET, DataSet modifications are not supported with IEC 61850 device	Either check if static DataSet can be used. Set DS_CONF_OP to "FALSE" and ensure your tag block sequence is the same as your tag order in the existing DataSet. Another possibility is to use "LIST" mode instead of DATASET mode.
		TIMEOUT or SAMPLE_T are not set properly	Check the TIMEOUT value. It's the time the block waits for a successful response to its requests. It should be at minimum 1000 ms. Running in LIST mode it should be at minimum 2000 ms. Check SAMPLE_T. It should represent the current scan time / 100.
		IEC61850_COM block is placed in more than one cyclic OBs	Ensure the IEC61850_COM block is placed only in one OB (e.g., OB35).
2.	IEC_RD_XX tag QBAD = 1	Invalid IEC 61850 address string	Check the address of your tag with the IEC 61850 browser. Remember that in Step 7, the additional sign "\$" has to be replaced with "\$\$". Further keep in mind, for IEC_ALARM, IEC_REPORT and IEC_CTRL only the address including the IEC 61850 object name has to be set. For all others the IEC 61850 attribute has to be added to the address.
		IEC 61850 Communication is not established	Check the IEC61850_COM block, if running in QSTATE = "ZYKLCOM". If not, check its parameters.
		IEC61850_COM IED_NAME does not equal IEC 61850 device name	Check IED name of IEC 61850 device.
3.	IEC61850_COM QBLOCKS and QTAGS output does not fit to tag configuration	No RESET executed	Changes to the tag configuration, like address change, adding tag blocks, removing tag blocks are taken over with executing a RESET. Execute a RESET.
4.	PLC Scan time exceeded	Tag blocks are not executed in the optimal OB	To get a better performance of the PLC system, the tag blocks can be put in a lower numbered OB (e.g., OB34, OB33, OB32) with bigger scan times (e.g., 200 ms, 500 ms, 1 s). With this method you can optimize your PLC system.

No	Problem	Possible Reason	Solution
5.	Reporting does not work, no WinCC messages	WinCC settings are not correct	Please check, if your OS is updated and works with your PLC program. Check if all connections to the PLC are established and working. IEC_REP block should update the timestamp after the report was generated. (Compare IEC_REP timestamp with tag timestamp get by IEC 61850 browser).
		RP_EN_QB is set to "1"	Something failed enabling Reporting with the Report Control block. Check if your RP_DS_ADDR0/1 match with DataSet address get by IEC 61850 browser.
		Reporting was not activated with DIGSI System Configurator (using a SIPROTEC)	Please follow up the procedure described in Appendix A.1.1 Report configuration with DIGSI if using a SIPROTEC device. If using a different device, follow up the manual instructions to add a Report application.
6.	"ZYKLCOM" state is reached, but IEC61850_COM block does not keep this state; communication gets instable.	IEC61850_COM TIMEOUT too small	Increase TIMEOUT value.

7. Appendix

7.1. Support Agreement

Information about Support Agreement can be found inside SIOS

<https://support.industry.siemens.com/cs/ww/de/view/109800036>

7.2. Service and support

Industry Online Support

Do you have any questions or need assistance?

Siemens Industry Online Support offers round the clock access to our entire service and support know-how and portfolio.

The Industry Online Support is the central address for information about our products, solutions and services.

Product information, manuals, downloads, FAQs, application examples and videos – all information is accessible with just a few mouse clicks:

support.industry.siemens.com

Technical Support

The Technical Support of Siemens Industry provides you fast and competent support regarding all technical queries with numerous tailor-made offers

– ranging from basic support to individual support contracts.

Please send queries to Technical Support via Web form:

siemens.com/SupportRequest

SITRAIN – Digital Industry Academy

We support you with our globally available training courses for industry with practical experience, innovative learning methods and a concept that's tailored to the customer's specific needs.

For more information on our offered trainings and courses, as well as their locations and dates, refer to our web page:

siemens.com/sitrain

Service offer

Our range of services includes the following:

- Plant data services
- Spare parts services
- Repair services
- On-site and maintenance services
- Retrofitting and modernization services
- Service programs and contracts

You can find detailed information on our range of services in the Industry Mall mall.industry.siemens.com.

7.3. Industry Mall



The Siemens Industry Mall is the platform on which the entire siemens Industry product portfolio is accessible. From the selection of products to the order and the delivery tracking, the Industry Mall enables the complete purchasing processing – directly and independently of time and location:
mall.industry.siemens.com

7.4. Related literature

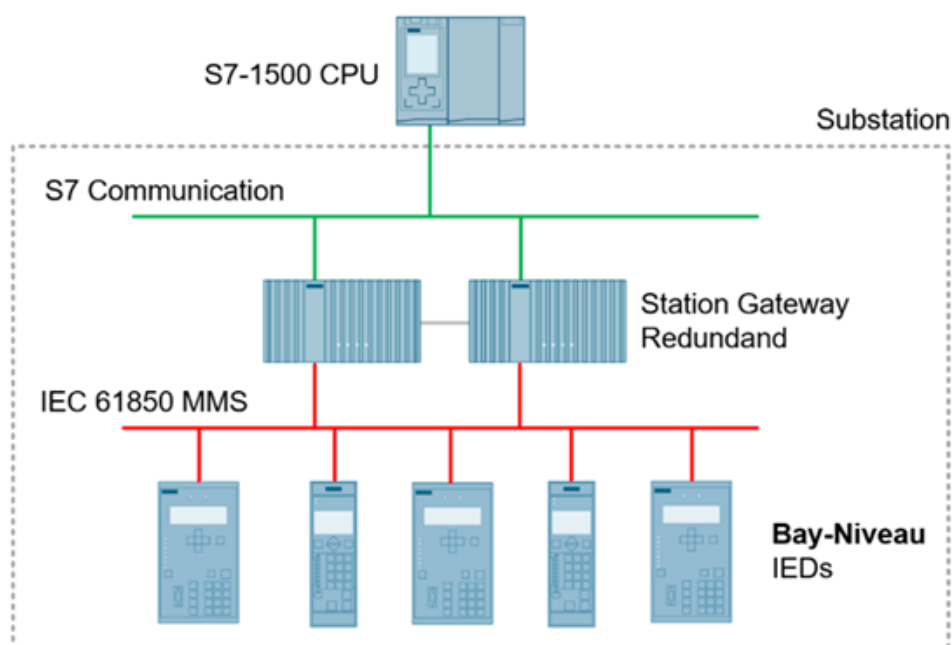
Table 7-1

Topic	
\1\	Siemens Industry Online Support https://support.industry.siemens.com
\2\	Download page of this entry https://support.industry.siemens.com/cs/ww/en/view/109779565
\3\	

7.5. Related products

7.5.1. Station Gateway for TIA Portal

The TIA Station Gateway based on the IPC227E offers a high-performance and flexible option for connecting up to 50 manufacturer-independent IEC 61850 servers (e.g. protection devices).



The following transfer methods are supported:

- **Data reporting (read)**
Messages/alarms via IEC 61850 reporting. The TIA Station Gateway buffers messages with time stamps from the protection relays and transfers them to the SIMATIC S7 15xx CPU.
- **Cyclical data (read)**
The data types BOOL, REAL, INT, DINT can be transmitted cyclically from the protection relays to the outputs of the function blocks.
- **Commands (write)**
Binary commands and interlocks are transmitted from function blocks on a SIMATIC S7-15xx CPU to the protection relays.

Furthermore the following functionalities are supported:

- **TIA Station Gateway redundancy:** two TIA Station Gateways can be used as a single redundant, highly available gateway
- **IEC 61850 server time stamps** can be used inside TIA Portal

The communication between SIMATIC S7-15xx and TIA Station Gateway is done via S7 connection. Therefore the TIA Station Gateway requires a SIMATIC S7-1500, which supports BSEND/BRECV.

Additional links for Station Gateway for TIA Portal

Delivery Release: <https://support.industry.siemens.com/cs/ww/en/view/109825352>

For more information about this topic, please contact:

<mailto:function.blocks.industry@siemens.com>.

7.5.2. SIMATIC Energy Suite

SIMATIC Energy Suite - Integrated energy management in the TIA Portal

As an option for the TIA Portal, SIMATIC Energy Suite links energy management with automation and introduces load management and thus flexibility of the energy consumption in addition to energy transparency in production and infrastructure. In order to create transparency, energy data is recorded in the Energy Suite, archived and provided in standardized form in the PLC. Energy data can then be visualized and used in your automation program (e.g. for monitoring). Energy data can then be visualized and used in the automation program (e.g. for monitoring).

This data can then be used for load management; the following load management modules are currently available:

Peak load management:

To avoid expensive load peaks, through priority-based switching of energy consumers, generators, and storage units.

Base load management:

A monitoring component that ensures that the power consumption does not exceed or fall below the user-defined (upper and lower) power limits.

Simple parameterization and automatic generation of the PLC program and the related visualization significantly reduce the project planning effort.

1. Overview of the range of functions

The following properties characterize SIMATIC Energy Suite:

Optional extension in the TIA Portal and therefore integrated into the automation

Energy transparency through energy data collection

Configuration of electrical and non-electrical media (e.g. kWh, m³, kg, unit)

Archiving and basic reporting in the data log of WinCC Professional or WinCC Unified

Flexible consumption through load management

Priority-based load management for energy consumers, producers and storages

Integration in the PLC ensures high flexibility and high reliability

Simple and intuitive configuration instead of complex, manual programming

Automatic generation of the PLC energy program for S7-1500 controllers

Integration of data from different protocols possible based on PLC based integration (Profinet, OPC, Modbus, IEC 61850, etc...)

-

Generation of Energy Suite screens with SiVArc (no SiVArc license required)

New in V19:

Energy Efficiency Monitor for Machines

Full integration of Energy Efficiency Monitor for Machines into the Energy Suite

Generation of the visualization of the energy efficiency monitor for machines with SiVArc (no additional SiVArc license required)

Automatic program generation of the PLC program for S7-1500 controllers

Integration of Modbus TCP devices

Extended support of Modbus TCP devices (Siemens and third-party) by automatic driver generation for communication

Templates for multiple use of Modbus-TCP devices of one type

Representation of extended energy data for Modbus-TCP devices in WinCC Unified

Support of redundancy concepts with the Energy Suite

S7-1500 R/H supports the Energy Suite

WinCC Unified

Improved user experience by revision of WinCC Unified visualization

Evaluation of energy data and automatic creation of reports with the Energy Suite Export Tool for WinCC Unified

2. Design

SIMATIC Energy Suite consists of the following components:

Engineering components (TIA Portal)

Adds new, integrated editors to the TIA Portal for configuration of the energy measurement and load management objects

Includes S7 program generator for automatic generation of the complete S7 program for acquisition, processing, buffering and archiving (WinCC Professional/WinCC Unified or SIMATIC memory card) of the energy values relevant to billing (power rating and energy) as well as all load management functions of the PLC

WinCC Runtime components (toolbox for WinCC Professional or WinCC Unified)

Export tool for creating basic energy reports of the energy data archived periodically in WinCC Professional or WinCC Unified to Excel

Transfer tool for buffered communication of the periodic energy values relevant to billing from S7-1500 CPUs to the data log of WinCC Professional or WinCC Unified.

Energy Efficiency Monitor for Machines

Designed for machine manufacturers and enables machines to be equipped with energy transparency functions, thereby completing the functional scope of the Energy Suite on the machine.

The following functions are provided:

Production-related and standardized determination of energy consumption and efficiency ratios in machines according to VDMA 34179 (e.g. kWh/unit, m³/unit)

Up to 10 types of energy per machine (e.g. electric, compressed air, water)

Up to 8 individually definable operating states (e.g. production, standby, off)

Easy integration into machine control system (S7-1200/1500) and on-site visualization of the efficiency status

Automatic long-term measurements (e.g. batch, shift)

Preparation of an efficiency report (.csv) for detailed evaluation and documentation

Additional links for Energy Suite:

- Delivery Release: <https://support.industry.siemens.com/cs/us/en/view/109822237>
 -
- Energy Management Media Center:
<https://support.industry.siemens.com/cs/us/en/view/109765100>
 -
- Energy Management Video platform:
<https://www.siemens.com/global/en/products/automation/industry-software/automation-software/energymanagement/simatic-energy-management-media-center.html>

7.6. Contact information

Should you have any questions concerning the software application, please refer to the Industry Sector Technical Support.

Department

Siemens AG

Industry Online Support

Phone

+49 (0) 721 667 1210 (Monday to Friday, 09:00 am to 05:00 pm CET/CEST)

Email

<mailto:function.blocks.industry@siemens.com>

Internet

<https://support.industry.siemens.com/>

Thank you for using one of the above-mentioned contacts to ensure your inquiry is registered and can be processed.

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